

FPT SCHOOL OF BUSINESS
& TECHNOLOGY

MASTER THESIS ON SOFTWARE ENGINEERING

HYBRID RETRIEVAL AND LEGAL-DOMAIN ADAPTATION FOR VIETNAMESE LEGAL INFORMATION RETRIEVAL

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Acknowledgements and Declaration of Authorship

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Declaration of Authorship

I declare that this thesis is my own work and has been prepared in accordance with accepted standards of academic integrity. All sources and materials used in this research have been appropriately acknowledged and cited.

The work presented in this thesis has not been submitted, either in whole or in part, for any other degree or academic qualification at this or any other institution.

I take full responsibility for the content, accuracy, and academic integrity of this thesis.

Nghiem Phan Thanh Tuan – 2026

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List of Abbreviations

Terminology follows the retrieval literature and the task definitions introduced in Chapters 3–4 [1, 2, 3, 4].

Abbreviation	Meaning	Explanation
ALQAC	Automated Legal Question Answering Competition	A benchmark and competition for evaluating legal question answering and information retrieval systems.
BCA / MPS	Bộ Công an / Ministry of Public Security collection	A legal question–document collection derived from public questions and published answers associated with Vietnam's Ministry of Public Security.
BM25	Best Matching 25 lexical ranking function	A term-based retrieval method that ranks documents according to query-term matches, term frequency, and document length.
BGE-M3	BGE-M3 multilingual embedding model	A multilingual embedding model used in this thesis as a dense retrieval component.
CV	Cross-validation	An evaluation procedure that partitions data into subsets for model selection and performance estimation.
E5	E5 text-embedding model family	A family of dense embedding models; this thesis evaluates the `multilingual-e5-base` variant.
FT / PT	Fine-tuned / Pretrained	Distinguishes a task-adapted model from its original pretrained version.
IR	Information Retrieval	The task of retrieving and ranking documents that are relevant to a user's information need.
MAP	Mean Average Precision	A ranking metric that averages precision across relevant results and then across queries.
MRR	Mean Reciprocal Rank	A ranking metric based on the reciprocal rank of the first relevant result for each query.
NDCG	Normalized Discounted Cumulative Gain	A ranking metric that rewards relevant documents appearing higher in the ranked result list.

Continued on next page

Abbreviation	Meaning	Explanation
OOF	Out-of-fold	Predictions or results produced for samples excluded from the corresponding fitting fold.
RAG	Retrieval-Augmented Generation	An architecture that retrieves external information before or during text generation.
RQ	Research Question	A specific question that the thesis investigates empirically.
SD	Standard Deviation	A statistical measure describing dispersion around the mean.

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Chapter 1

Introduction

1.1 Research Context

Retrieval-Augmented Generation (RAG) is a technique that grounds language model outputs in external documents retrieved from a target corpus [4]. In legal question answering, retrieval precision is critical: an answer, regardless of how fluent, remains unreliable if the cited legal provisions fail to support it. Accordingly, this thesis centers on retrieval effectiveness. Given a Vietnamese legal query and a corpus of legal documents, the goal is to rank provisions such that relevant articles appear at the top of the retrieved results.

Retrieving Vietnamese legal documents poses several distinct challenges. User queries typically rely on everyday language, whereas statutory provisions are framed in formal legal terminology. Furthermore, articles addressing similar domains often dictate divergent conditions, obligations, or procedural requirements; thus, topical or lexical similarity alone does not establish legal relevance. Collection completeness also sets an upper bound on performance, as unindexed provisions cannot be retrieved. Additionally, misalignments between reference annotations and document identifiers can distort benchmark metrics. A rigorous evaluation must therefore account for ranking quality, corpus coverage, and mapping consistency.

To establish a baseline, this thesis employs BM25, which ranks documents via lexical matching a vital capability given that legal queries frequently contain explicit statutory identifiers, document numbers, or standard phrasing [5, 6]. In contrast, dense retrieval leverages embedding similarity to match queries and documents across lexical gaps. We evaluate two representative pretrained models: BGE-M3 and multilingual-E5-base (E5) [2, 3], along with hybrid configurations (BM25+BGE-M3 and BM25+BGE-M3+E5) that combine lexical and dense retrieval scores.

We evaluate these approaches on benchmark Vietnamese legal datasets using established relevance annotations. Specifically, we investigate whether BM25+BGE-M3 improves upon the BM25 baseline, whether integrating E5 yields further gains, and whether fine-tuning BGE-M3 enhances standalone retrieval and sustains improvements within hybrid pipelines. Through these empirical evaluations, this thesis clarifies the effectiveness of hybrid retrieval strategies for Vietnamese legal information retrieval.

1.2 Research Gap and Approach

Prior research provides a basis for the methods evaluated in this thesis. Rosa et al. report that BM25 is a competitive baseline for legal case retrieval [6]. Chen et al. and Wang et al. present BGE-M3 and multilingual E5 and evaluate their embedding models on multilingual benchmarks [2, 3]. Bruch et al. analyze fusion of lexical and semantic retrieval scores, including the effects of normalization and parameter selection [14]. These findings motivate comparisons between lexical retrieval, dense retrieval, and their combinations.

The scope of that evidence also matters. Legal case retrieval differs from retrieving statutory articles for Vietnamese questions. In heterogeneous retrieval evaluation, BEIR identifies limitations in the generalization of dense retrievers across datasets [17]. Zhan et al. show that training with hard negatives can improve dense retrieval on their evaluated benchmarks [16]; their findings concern dense-retriever training rather than the effect of fine-tuning BGE-M3 within the hybrids evaluated here. Therefore, the literature provides a rationale for testing these methods, while their effectiveness on the selected Vietnamese legal collections remains an empirical question.

The research gap addressed here is the evidence needed for two specific decisions: whether adding E5 improves BM25+BGE-M3, and whether fine-tuning BGE-M3 improves both standalone and hybrid retrieval. The thesis tests these questions under defined corpus conditions and separates model selection from evaluation on a separate test set. This is an empirical contribution rather than a new retrieval algorithm. Chapter 2 reviews the supporting literature, and Chapter 4 specifies the comparisons and evaluation procedures.

The approach has three parts. First, the overall retrieval benchmark evaluates ALQAC and Zalo using only the reference article text supplied with each dataset. All retrieval methods within a dataset use the same questions, corpus, and recorded relevance targets.

Second, a Bộ Công an (BCA) dataset is constructed from published public-service questions and legal citations in their answers. The evaluation examines whether the cited provisions are present in the corpus and identifies retrieval failures. The evaluation result is reported separately due to differences in data source from ALQAC and Zalo.

Third, the BGE-M3 embedding model is fine-tuned using legal question article pairs and evaluated in standalone and hybrid retrieval. BM25 and E5 remain unchanged, and fusion weights are held fixed when comparing pretrained and fine-tuned BGE-M3 within each hybrid. The analysis distinguishes development scores from results on test set queries.

1.3 Research Objectives

The main objective is to evaluate hybrid retrieval and fine-tuning of the BGE-M3 embedding model for Vietnamese legal information retrieval. The thesis also constructs a new dataset

BCA for evaluation, the goal of which is to provide a more comprehensive assessment of retrieval performance in the Vietnamese legal domain. The thesis has four specific objectives:

1. Compare BM25, pretrained BGE-M3, pretrained E5, and BM25+BGE-M3 on ALQAC and Zalo datasets. BM25's evaluation scores serve as the baseline.
2. Determine whether adding E5 improves retrieval by comparing BM25+BGE-M3+E5 with BM25+BGE-M3.
3. Evaluate whether fine-tuning BGE-M3 improves the performance in BM25+BGE-M3 and BM25+BGE-M3+E5.
4. Construct and evaluate the BCA collection from published public-service questions and legal citations in their answers. Benchmark retrieval performance, examine the differences between public questions with Zalo and ALQAC.

Retrieval effectiveness is measured primarily by NDCG@10. Hit@10, MRR@10, and MAP@10 provide additional measures of retrieval performance within the top ten results. Comparisons are reported separately for each dataset and corpus condition to establish the scope of any observed improvement.

1.4 Research Contributions

The thesis contribution is an empirical comparison of retrieval methods performance including BM25, BGE-M3, E5, BM25+BGE-M3, and BM25+BGE-M3+E5 on ALQAC, Zalo, and BCA. The evaluation measures retrieval performance on datasets ALQAC, Zalo, and BCA. The results show that hybrid retrieval improves NDCG@10 over BM25 in most evaluated conditions, while adding E5 provides little to no additional benefit (Tables 5.16 and 5.17).

The second contribution is an evaluation of fine-tuning the BGE-M3 embedding model for standalone and hybrid retrieval. The first experiment selects the fine-tuned model based on hybrid retrieval performance on development data. The second experiment selects the model based on standalone dense retrieval performance on development data. The result shows fine-tuning improves metric NDCG@10 but does not produce a meaningful improvement in either hybrid when fusion weights are the same (Tables 5.11 and 5.12).

The third contribution is the construction and evaluation of the BCA retrieval collection from published public-service questions and legal citations in their answers. The analysis examines the availability of cited provisions in the corpus and dependence among questions linked by shared legal references. The analysis also examines the sortfall failures of the BCA collection (Table 5.3).

1.5 Scope of the Thesis

The thesis evaluates the ranking of legal articles for Vietnamese questions on ALQAC, Zalo, and BCA. It compares BM25, BGE-M3, E5, and their hybrid configurations, and examines the effect of fine-tuning BGE-M3. The evaluation covers retrieval performance, corpus coverage, and the mapping of cited provisions to indexed documents. Answer generation, the quality of legal advice, user satisfaction, the currency of legal documents, and deployment performance are outside its scope.

The evaluation uses Vietnamese questions and legal texts. BGE-M3 and E5 are multilingual models, but the experiments do not assess retrieval in other languages or legal systems. Conclusions are limited to the evaluated datasets, corpus conditions, model configurations, and fusion weights. Applying the findings to other collections or downstream RAG systems requires further evaluation.

For BCA dataset the data is collected from publicly available questions and answers on the Ministry of Public Security's website on the date of collection. Many questions are refrencing private legal documents, which is inaccessible to the public. The evaluation does not attempt to access or retrieve any private legal documents, and the analysis is limited to publicly available information.

1.6 Thesis Organization

Chapter 1: Introduction presents Vietnamese legal information retrieval and the research gap addressed by the thesis. It states the research objectives, contributions, and scope.

Chapter 2: Literature Review reviews prior work on legal retrieval, BM25, multilingual embedding models, hybrid retrieval, and fine-tuning. It also discusses retrieval benchmarks and evaluation methods to establish the basis for the thesis's experimental comparisons.

Chapter 3: Theoretical Background explains lexical and dense retrieval, score normalization and fusion, and contrastive training. It defines relevance, evaluation metrics, and the principles used to account for dependence among queries when interpreting results.

Chapter 4: Research Methodology describes dataset and corpus construction, including the BCA collection, and the retrieval pipeline. It specifies fusion weight selection, BGE-M3 fine-tuning, and the procedures for model selection and evaluation.

Chapter 5: Experiments and Evaluation describes the experimental conditions and benchmark evaluation procedure. Its results and analysis cover the BCA collection, BGE-M3 fine-tuning, and the overall comparison of individual and hybrid retrieval methods, including the contribution of E5.

Chapter 6: Conclusion and Future Work summarizes the findings and contributions

in relation to the research objectives. It discusses the limitations of the evaluation and outlines directions for future work.

Chapter 2

Literature Review

This chapter reviews prior work on legal retrieval, multilingual embeddings, and hybrid fusion. It identifies the research gap addressed by this thesis and positions its contributions relative to the literature.

Figure 2.1 shows the position of retrieval in a Retrieval-Augmented Generation (RAG) system. The retrieval component searches the document sources and supplies information to the generator to support answer generation. If the retriever fails to return relevant information, the generator cannot produce a correct answer. Therefore, retrieval performance is a prerequisite for accurate answer generation.

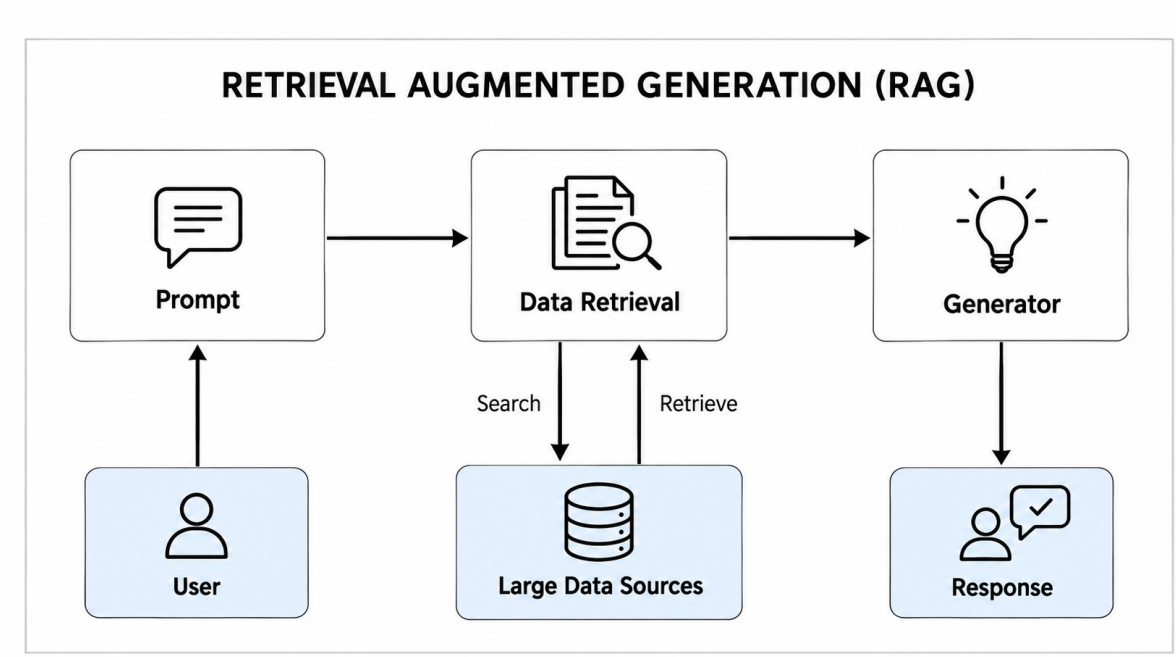


Figure 2.1: Position of retrieval in a Retrieval-Augmented Generation (RAG) system.

2.1 Legal Retrieval and Evidence

Legal information retrieval identifies and ranks legal materials relevant to a query. The retrieval unit may be a judgment, statute, article, or passage. The Competition on Legal Information Extraction and Entailment (COLIEE) separates retrieval tasks from entailment tasks, which assess whether legal texts support a given statement. Its case tasks

use Canadian case law, while its statute tasks use Japanese bar-examination questions [7]. Comparisons with Vietnamese article retrieval must therefore account for differences in the legal system, retrieval unit, relevance annotations, and evaluation metrics.

The Automated Legal Question Answering Competition (ALQAC) 2024 provides manually annotated Vietnamese legal questions and defines separate tasks for legal document retrieval and question answering. In the retrieval task, articles are relevant when they can be used to answer the question [8]. This thesis uses the available relevance annotations to assess how highly the recorded relevant articles are ranked.

Similarity in wording or topic alone does not establish relevance to a legal question. For example, a question about an application deadline may match an article concerning a different stage of the same procedure. In this hypothetical case, the article discusses the same topic but does not provide the required deadline. A relevant provision may also use different terms from those in the question. These possibilities motivate comparing lexical and dense retrieval using the same queries and relevance annotations.

The retrieval unit must also match the relevance annotations. An article may contain rules, exceptions, and references to other provisions, so splitting it into passages changes the context available to the retriever. When the annotations identify articles, retrieved passages must be mapped to their source articles before ranking performance is evaluated. Otherwise, passage retrieval and article retrieval would be assessed using different units.

Text preprocessing is another consideration in Vietnamese retrieval. PhoBERT uses word segmentation before subword tokenization and is evaluated on Vietnamese language-processing tasks [9]. Its preprocessing illustrates a model-specific requirement; its results do not establish that word segmentation improves the retrievers used in this thesis. For lexical retrieval, splitting text at spaces and segmenting it into words produce different vocabularies and term counts. For dense retrieval, preprocessing should follow the embedding model's documented requirements.

In the RAG workflow shown in Figure 2.1, retrieval supplies information to the generator. Lewis et al. combine a language model with retrieved passages for knowledge-intensive tasks [4]. This thesis evaluates the retrieval component by measuring article ranking against the available relevance annotations. Therefore, its results establish retrieval performance within those annotations, while any improvement in the accuracy of generated legal answers requires a separate evaluation.

2.2 Lexical Retrieval and BM25

Lexical retrieval ranks documents by matching query terms with indexed text. BM25 is a probabilistic ranking model that combines term frequency and document frequency with document length normalization [5]. Its scores therefore depend on both the terms in a

document and their distribution in the indexed corpus.

A BM25 submission achieved second place in the legal case retrieval task of COLIEE 2021 [6]. This finding provides empirical support for using BM25 as a baseline in legal retrieval. Since that task concerns case law, its results do not establish BM25's effectiveness for Vietnamese legal article retrieval. The latter requires evaluation on the Vietnamese collections studied here.

This thesis compares dense and hybrid retrieval with BM25 using the same queries, corpus, and relevance annotations within each evaluation condition. BM25's evaluation scores serve as the baseline. Section 3.2 explains the scoring formula, and Section 4.5 specifies the preprocessing and retrieval implementation used in the experiments.

2.3 Dense Retrieval and Multilingual Embeddings

Dense retrieval represents queries and documents as embeddings and ranks documents by vector similarity. Sentence-BERT produces sentence embeddings that can be compared using cosine similarity [10]. Dense Passage Retrieval trains separate query and passage encoders for open-domain question answering [11]. Independent encoding allows document embeddings to be computed before a query is submitted and reused during retrieval.

BGE-M3 supports multilingual retrieval through dense, sparse, and multiple-vector representations [2]. Multilingual E5 provides embedding models in small, base, and large variants, trained using multilingual text pairs and labeled datasets [3]. This thesis evaluates BGE-M3's dense representations and multilingual-E5-base (E5). The results therefore apply to these configurations, rather than to every retrieval mode or model variant described in the source papers.

Single-vector retrieval represents each query and document with one vector. ColBERT instead retains contextualized token embeddings and computes relevance through interactions between query and document tokens after encoding [12]. This approach provides an alternative to single-vector retrieval. ColBERT is included here as related work and is not evaluated in the experiments.

Multilingual-E5-base is included to test whether it complements BGE-M3 in semantic retrieval. The models use different training procedures and embedding methods [2, 3]. The hypothesis is that these differences may produce complementary rankings, allowing E5 to improve retrieval for some questions whose relevant articles are ranked poorly by BGE-M3. Whether this improves hybrid retrieval is tested by comparing BM25+BGE-M3+E5 with BM25+BGE-M3.

The experiments also compare pretrained BGE-M3 and E5 with BM25 to assess standalone retrieval on the Vietnamese legal collections, and BM25+BGE-M3 with BM25 to assess

hybrid retrieval. These comparisons distinguish the effect of combining lexical and dense retrieval from the effect of adding E5 to an existing hybrid.

2.4 Hybrid Retrieval and Score Fusion

Hybrid retrieval combines the outputs of different retrievers to produce a single ranking. Reciprocal Rank Fusion combines rankings using document positions rather than the original retrieval scores [13]. Weighted score fusion instead combines the component scores according to assigned weights. An analysis of lexical and semantic retrieval examines how score normalization and parameter selection affect fusion performance [14]. These findings motivate specifying both the normalization method and the procedure for selecting fusion weights.

The hypothesis is that lexical and dense retrieval may provide complementary rankings. A relevant article ranked poorly by one retriever may be ranked highly by another. Adding E5 to BM25+BGE-M3 tests whether another dense retriever improves a system that already combines lexical and semantic information. Its standalone performance does not establish that additional benefit.

These studies motivate evaluating whether combining lexical and dense retrieval improves Vietnamese legal article ranking, and whether an additional dense retriever provides further benefit. Section 3.4.3 explains the fusion calculation, while Section 4.6 specifies the experimental procedure. Reciprocal Rank Fusion is discussed as related work but is not evaluated in the experiments.

2.5 Fine-Tuning for Legal Retrieval and Hard Negatives

Contrastive training teaches a retrieval model to give relevant passages higher similarity scores than passages labeled as irrelevant [11].

Hard negatives are passages that a retriever ranks highly even though they are labeled as irrelevant. They help train the model to distinguish relevant passages from similar but incorrect matches. Experiments show that mixing hard negatives with randomly selected negatives, or selecting new hard negatives during training, can improve dense retrieval [16].

These findings motivate fine-tuning a dense retriever, and the experiments test whether fine-tuning BGE-M3 can improve retrieval on the Vietnamese legal collections.

2.6 Research Gap and Positioning

BGE-M3 and multilingual E5 provide pretrained embedding models for multilingual retrieval [2, 3]. Research on lexical and semantic score fusion examines the effects of normalization and parameter selection [14], while hard-negative training strategies have improved dense retrieval on benchmark datasets [16]. These studies provide a basis for evaluating multilingual embeddings, score fusion, and embedding model fine-tuning on Vietnamese legal collections.

The research gap addressed by this thesis is the evidence needed to assess these methods under the selected Vietnamese legal datasets. The evaluation compares BM25, BGE-M3, and E5 in standalone and hybrid retrieval, and tests whether fine-tuning BGE-M3 with hard negatives improves retrieval.

The contribution is a controlled empirical comparison of established retrieval methods, together with the construction and evaluation of the BCA collection. The BCA analysis examines the availability of cited provisions and dependence among questions linked by shared legal references. Chapter 4 specifies the datasets and evaluation procedures, and Chapter 5 reports the results within those conditions.

2.7 Chapter Summary

This chapter reviewed legal retrieval, BM25, multilingual embeddings, score fusion, and fine-tuning with hard negatives. The reviewed studies motivate evaluating these methods on Vietnamese legal collections, with separate comparisons for improvements over BM25, the contribution of E5, and the effect of BGE-M3 fine-tuning in standalone and hybrid retrieval. Chapter 3 presents the theoretical background and evaluation metrics used in these comparisons.

Chapter 3

Theoretical Background

This chapter presents the theoretical foundations of retrieval models, score fusion, model training, and evaluation metrics used in the thesis. Chapter 4 describes how these methods are implemented and evaluated.

3.1 Queries, Articles, and Relevance

Information retrieval ranks indexed documents according to their relevance to a query [1]. In this thesis, the query is a Vietnamese legal question, and the retrieval unit is a legal article.

The corpus contains laws (Luật), decrees (Nghị định), and circulars (Thông tư). A document may be organized into parts (phần), chapters (chương), sections (mục), subsections (tiểu mục), articles (điều), clauses (khoản), and points (điểm), although not every document uses all these levels [21]. Each article belongs to one legal document and may contain several clauses and points.

Chunking divides document text into units for indexing and retrieval. The choice of unit can affect retrieval performance [22]. This thesis uses articles as the retrieval units. For documents collected in raw HTML format, the extracted text is separated at article boundaries. Each article record contains its text, document identifier, and article identifier. Clauses and points are kept within their source article. Section 4.2 describes corpus construction and the Zalo reference-text fallback when extraction is unavailable.

Chunking can affect recall because it determines the content available to the retriever before encoding. If a chunk incorrectly combines three articles, one vector represents their combined content. This may make a specific provision harder to retrieve and associate text with the wrong article identifier. Improving the encoder does not repair missing text or incorrect identifiers in the corpus. Article boundaries and the match between text and identifiers must therefore be checked when investigating retrieval errors.

Example of an article-level record. The following example contains two article excerpts from the supplied sample for Decree No. 340/2025/NĐ-CP. The title and article texts are shortened for display, with ellipses marking omitted text. The example illustrates the record format; the complete sample retains each article's clauses and points.

```
{
  "law_id": "340/2025/NĐ-CP",
  "title": "NGHỊ ĐỊNH SỐ 340/2025/NĐ-CP ...",
  "url": "https://vbpl.vn/van-ban/chi-tiet/nghi-dinh-so-340-2025-nd-cp-quy-dinh-ve-xu
  ↪  -phat-vi-pham-hanh-chinh-trong-linh-vuc-tien-te-va-ngan-hang--187101",
  "articles": [
    {
      "id": "1",
      "text": "Điều 1. Phạm vi điều chỉnh\n1. Nghị định này quy định các hành vi vi
      ↪  phạm hành chính ... trong lĩnh vực tiền tệ và ngân hàng.\n2. Vi phạm hành
      ↪  chính trong lĩnh vực tiền tệ và ngân hàng bao gồm: ..."
    },
    {
      "id": "2",
      "text": "Điều 2. Đối tượng bị xử phạt vi phạm hành chính\n1. Nghị định này áp
      ↪  dụng đối với tổ chức, cá nhân có hành vi vi phạm hành chính trong lĩnh vực
      ↪  tiền tệ và ngân hàng.\n2. Tổ chức quy định tại khoản 1 Điều này gồm: ..."
    }
  ]
}
```

The fields `law_id`, `title`, and `url` describe the legal document. Each entry in `articles` is a separate chunk with its own `id` and `text`. Article 1 and Article 2 therefore receive separate embeddings. Together, `law_id` and the article `id` identify each retrieval unit.

Relevance targets identify the documents or articles recorded as relevant to each query. ALQAC and Zalo use the available dataset annotations. For BCA, the question is used as the query, and citations in its published answer are extracted to form relevance targets. Document names and article numbers are normalized and matched to corpus identifiers. The answer text is used for target extraction and is not supplied to the retriever. Table 4.2 illustrates this process.

An article-level target specifies a document and an article within it. A law-level target specifies only the document. In the dataset, "law-level" also covers decrees and circulars. Duplicate targets are removed, and each target receives evaluation credit at most once. Retrieving several articles from one document therefore does not multiply credit for a single law-level target. Section 3.6 defines the matching rules and evaluation metrics.

Retrieval returns a list of article identifiers ordered by retrieval score. These identifiers link the results to the article text and source document. A RAG generator can use the retrieved text to produce an answer, while this thesis evaluates the article ranking. Corpus coverage measures target availability, and ranking performance measures target positions.

3.2 Lexical Scoring with BM25

3.2.1 Term Frequency and Inverse Document Frequency

Term frequency and inverse document frequency (TF-IDF) are combined to weight indexed terms. A term receives a higher weight when it occurs frequently in an article but appears in relatively few documents across the corpus [1]. This illustrates how lexical weighting combines term occurrence with term rarity. The experiments use BM25 as the lexical baseline; TF-IDF is included here as background.

3.2.2 BM25

BM25 is a popular lexical scoring function that combines term frequency, inverse document frequency, and document length normalization. For a query q and an article d, a common scoring formula sums the contributions of query terms t [5]:

$$\mathrm{BM25}(q,d) = \sum_{t \in q} \mathrm{IDF}(t) \frac{f(t,d)(k_1+1)}{f(t,d) + k_1\left(1 - b + b|d|/\,\mathrm{avgdl}\right)}.$$

The quantities in the formula are:

- $f(t,d)$: the number of occurrences of term t in article d.
- $|d|$: the article length, measured in indexed tokens.
- avgdl: the average article length in the indexed corpus.
- $\mathrm{IDF}(t)$: the inverse document frequency weight of term t.

The parameter k_1 controls term frequency saturation: repeated occurrences of a term contribute progressively less to its score. The parameter b, between zero and one, controls document length normalization. Setting $b = 0$ removes length normalization, while $b = 1$ applies full length normalization [1].

BM25 scores depend on the indexed text. Text extraction can change term counts, article lengths, and corpus statistics, so different corpus representations may produce different scores. A query term absent from an article contributes zero to its BM25 score. Section 4.5 describes the lexical preprocessing used in the experiments.

BM25 relies on matching indexed terms. A relevant article expressed using different terms from the query may therefore receive a low score. The thesis compares BM25 with dense and hybrid retrieval to evaluate whether embedding similarity improves the ranking. BM25 scores determine article order, while evaluation metrics such as NDCG@10 measure ranking quality against the relevance annotations. The baseline is BM25's evaluation performance on the target collections.

3.3 Dense Retrieval and Vector Similarity

An embedding is a numerical vector that represents text. In a bi-encoder architecture, the query and article are encoded separately. Dense retrieval then ranks articles by the similarity between their embeddings and the query embedding. Sentence-BERT produces sentence embeddings for similarity comparison [10]. Dense Passage Retrieval trains query and passage encoders to retrieve relevant passages [11]. Article embeddings can be computed in advance and reused across queries.

Figure 3.1 illustrates separate query and document encoding, similarity scoring, and ranking.

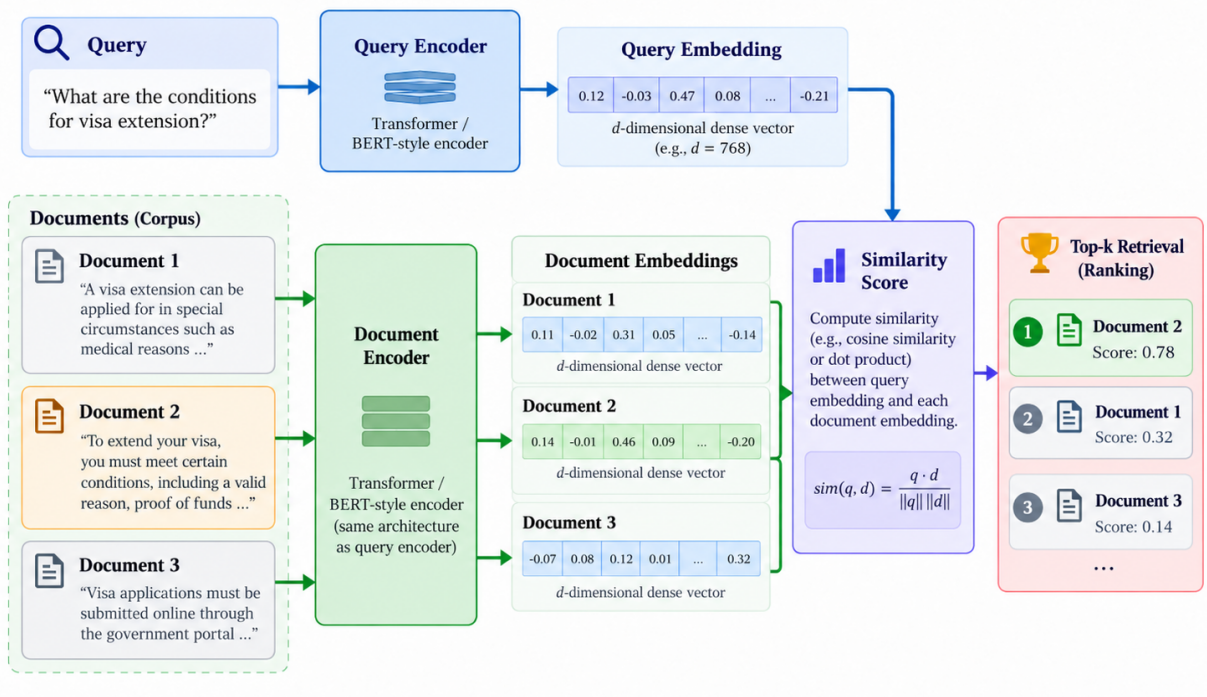


Figure 3.1: Illustrative examples of dense retrieval workflow. The query and document are encoded separately into embeddings, which are then compared using a similarity measure to rank the documents.

For a query q and an article d, u and v are their respective embedding vectors. Two common similarity measures are the dot product and cosine similarity:

$$s_{\text{dot}}(q, d) = u^\top v, \qquad s_{\cos}(q, d) = \frac{u^\top v}{\|u\|_2 \|v\|_2}.$$

The dot product $u^\top v$ sums the products of corresponding vector components. The quantities $\|u\|_2$ and $\|v\|_2$ are the vector lengths. Cosine similarity divides the dot product by these lengths and is defined for nonzero vectors. When both vectors are normalized to length one, the dot product equals cosine similarity.

A pooling operation produces one embedding from the model's token representations.

Input longer than the selected token limit is truncated, so the embedding may omit content that remains in the stored article. ColBERT retains multiple token embeddings for matching [12]. This thesis uses single-vector dense retrieval.

3.3.1 BGE-M3

BGE-M3 has three main features:

1. **Multi-Functionality**

   BGE-M3 supports three retrieval methods [24]:

   - **Dense Retrieval:** Uses dense vector representations to capture semantic meaning.
   - **Sparse Retrieval:** Uses sparse representations to match important keywords and lexical features.
   - **Multi-Vector Retrieval:** Uses multiple vectors with late interaction mechanisms, similar to ColBERT, to capture relevance through comparisons between query and document tokens.

2. **Multi-Linguality**

   - Supports more than 100 languages [2].
   - Enables cross-lingual retrieval, allowing a query to retrieve documents written in a different language [2].

3. **Multi-Granularity**

   - Handles text inputs of varying lengths, from short sentences to long documents [24].
   - Supports input sequences of up to 8,192 tokens for encoding short passages and longer documents [24].

This thesis uses BGE-M3's dense representations for retrieval and fine-tuning. The baseline experiments use a maximum input length of 512 tokens (Table 4.3). The hybrid systems combine BM25 scores with BGE-M3 dense similarity scores. BGE-M3's sparse and multiple-vector modes are outside the evaluated configurations.

3.3.2 Multilingual-E5-base

Multilingual E5 models are pretrained on multilingual text pairs and then fine-tuned on labeled datasets [3]. This thesis uses multilingual-E5-base, released as

`intfloat/multilingual-e5-base`. The model represents each input text as a vector with 768 components [23]. Its weights remain unchanged throughout the experiments.

E5 is included to test whether combining two dense retrievers improves hybrid retrieval. Its contribution is measured by comparing BM25+BGE-M3+E5 with BM25+BGE-M3 using the same queries, corpus, and relevance annotations within each evaluation condition.

3.4 Score Normalization and combination weights

3.4.1 Hybrid Retrieval

Hybrid retrieval combines scores or rankings from different retrievers. The aim is to improve article ranking by combining lexical matches with embedding similarity. The result depends on how the scores are normalized and combined [14]. This thesis evaluates BM25+BGE-M3 and BM25+BGE-M3+E5.

Scores can be combined for all articles in the corpus or only for candidates selected by the retrievers. In the first approach, each retriever scores every indexed article before the scores are combined. In the second, each retriever first selects its top candidates, and fusion ranks only the articles in those lists. An article missing from every candidate list cannot appear in the final results, regardless of the weights used. This thesis uses the first approach: each retriever scores the full corpus before fusion, as described in Section 4.5.

3.4.2 Score Normalization

The retrievers produce raw scores on different scales:

- **BM25:** The score depends on query term matches, term frequency, inverse document frequency, and article length. It has no fixed numerical range and is not normalized to $[0, 1]$. Its values depend on the query, corpus, and inverse document frequency convention [5].
- **BGE-M3:** The pipeline normalizes query and article embeddings to length one and computes their dot product. For nonzero vectors, this equals cosine similarity, whose theoretical range is $[-1, 1]$.
- **Multilingual-E5-base:** The pipeline uses the same normalized-vector dot product, so its theoretical score range is also $[-1, 1]$. Its scores are computed from E5 embeddings rather than BGE-M3 embeddings.

Section 4.5 specifies the score calculation used in the experiments. Although BGE-M3 and E5 share the same theoretical range, their scores can have different distributions across

articles. A raw score from one retriever is therefore not directly interchangeable with a score from another.

Vector normalization and score normalization serve different purposes. Vector normalization gives each embedding a length of one. Min-Max score normalization rescales each retriever's scores across the corpus for a particular query before weighted fusion [14]. For raw score $s_j(q, d)$ from retriever j, the form used in this thesis is

$$\widetilde{s}_j(q, d) = \frac{s_j(q, d) - \min_{x \in C} s_j(q, x)}{\max\left(\max_{x \in C} s_j(q, x) - \min_{x \in C} s_j(q, x), \epsilon\right)},$$

where C is the indexed corpus and $\epsilon > 0$ prevents division by zero. The minimum and maximum are computed separately for each query and retriever. The resulting scores lie in $[0, 1]$. If all articles receive the same raw score, their normalized scores are zero. Normalization preserves each retriever's score ordering; it changes the scale used for fusion. These scores are used to rank articles and are distinct from evaluation metrics such as normalized discounted cumulative gain. Section 4.6 specifies the implemented normalization procedure.

An alternative is z-score normalization:

$$Z_j(q, d) = \frac{s_j(q, d) - \mu_j(q)}{\sigma_j(q)},$$

where $\mu_j(q)$ and $\sigma_j(q) > 0$ are the mean and standard deviation of the retriever's scores over the corpus. Z-scores can be negative and are not restricted to the range $[0, 1]$. This method is included as background; the reported experiments use Min-Max normalization. Both methods depend on the articles included in normalization, so the normalization set must be stated.

3.4.3 Weighted Score Fusion

Weighted score fusion adds the normalized retrieval scores after multiplying each by a weight [14]. The weights control how much each retriever contributes to the combined score. For a query q and an article d, $\widetilde{s}_b$, $\widetilde{s}_g$, and $\widetilde{s}_e$ denote the normalized scores from BM25, BGE-M3, and E5, respectively.

For BM25+BGE-M3, the combined score is

$$S_{BG}(q, d) = w_b \widetilde{s}_b(q, d) + (1 - w_b)\widetilde{s}_g(q, d), \qquad 0 \leq w_b \leq 1,$$

Here, w_b is the BM25 weight, and $1 - w_b$ is the BGE-M3 weight.

For BM25+BGE-M3+E5, the score becomes

$$S_{BGE}(q, d) = w_b \widetilde{s}_b(q, d) + w_g \widetilde{s}_g(q, d) + w_e \widetilde{s}_e(q, d),$$

The weights w_b, w_g, and w_e correspond to BM25, BGE-M3, and E5. They are nonnegative and satisfy $w_b + w_g + w_e = 1$. Articles are ranked by the combined score in descending order. Section 4.6 specifies the fusion procedure used in the experiments.

Example of Score Fusion. Consider the legal question:

> *"Is an employee entitled to severance allowance after terminating an employment contract?"*

BM25 and BGE-M3 independently assign retrieval scores to candidate legal articles. Their scores are normalized separately over the indexed corpus using the Min-Max procedure described above. The normalized scores are then combined using weights of 0.4 for BM25 and 0.6 for BGE-M3:

$$S_{BG}(q, d) = 0.4\widetilde{s}_b(q, d) + 0.6\widetilde{s}_g(q, d).$$

Table 3.1 shows hypothetical normalized scores for three candidate articles from the larger corpus. The article descriptions and weights are used only to illustrate the calculation.

Table 3.1: Illustrative BM25+BGE-M3 score fusion for three legal articles.

Legal article	**BM25**	**BGE-M3**	**Fused score**	**Final rank**
A: Severance allowance	0.90	0.95	**0.93**	**1**
B: Unemployment benefits	0.85	0.60	0.70	2
C: Contract termination	0.40	0.80	0.64	3

For Article A, the fused score is

$$S_{BG}(q, A) = 0.4 \times 0.90 + 0.6 \times 0.95 = 0.93.$$

The articles are ranked by fused score in descending order. Article A ranks first because both models assign it high scores. The example illustrates how fusion combines lexical matching from BM25 and embedding similarity from BGE-M3 into a single ranking score. It demonstrates the calculation, rather than establishing which provision legally answers the question.

3.5 Fine-Tuning and Negative Sampling

Pretraining on legal corpora develops language representations, as investigated in LEGAL-BERT [15]. Retrieval fine-tuning instead uses labeled questions and relevant passages to train their similarity scores [11]. This section explains the training objectives and negative examples used to fine-tune the BGE-M3 embedding model.

3.5.1 Contrastive Learning

BGE-M3 is fine-tuned to give relevant articles higher similarity scores than articles used as negative training examples. For a question q, d^+ is one article labeled as relevant, and $N(q)$ is the set of negative training examples. The similarity between a question and an article is denoted by $s(q, d)$. A single-positive contrastive loss is [11]

$$\mathcal{L}(q) = -\log \frac{\exp(s(q, d^+)/\tau)}{\exp(s(q, d^+)/\tau) + \sum_{d^- \in N(q)} \exp(s(q, d^-)/\tau)}.$$

The loss decreases when the relevant article receives a higher score relative to the negative examples. The temperature $\tau > 0$ scales the similarity scores within the loss. Smaller values increase the effect of score differences.

A question may have several relevant articles among its training candidates. The multi-positive objective used in this thesis sums their exponentiated scores in the numerator and sums the exponentiated scores of all candidates in the denominator. This allows several relevant articles to contribute to the objective for one question.

The experiment that selects the model based on hybrid retrieval performance uses a single-positive loss. The experiment that selects the model based on standalone dense retrieval performance uses a multi-positive loss. Section 4.7 specifies the candidates and loss used in each experiment. A lower training loss must be followed by evaluation on a separate test set to assess retrieval performance.

3.5.2 Negative Sampling

Negative sampling selects articles that the model is trained to rank below the known relevant articles. Three common methods differ in how the examples are selected [11, 16]:

- **Random negatives:** Articles selected at random from the corpus.
- **In-batch negatives:** Articles from other training examples processed in the same batch. An article relevant to another question may be treated as a negative for the current question unless it is also identified as relevant to that question.

- **Hard negatives:** Articles ranked highly by a retriever but selected as negatives because they have no recorded positive label for the question.

An article without a positive label may still be relevant. Excluding known relevant articles from negative candidates reduces contradictory training labels but does not resolve missing annotations. The training procedure must also state how it treats other known relevant articles within the same batch. Section 4.7 describes the negative-selection rules and treatment of positive examples in the fine-tuning experiments.

3.5.3 Hard-Negative Mining

Hard-negative mining selects negative training examples from a retriever's rankings [16]. The procedure must specify the retriever, corpus, candidate ranks, and filtering rules so that the training examples can be reproduced.

In the experiment that selects BGE-M3 based on hybrid retrieval performance, negatives are selected from lexical and dense rankings. Known relevant articles, duplicate candidates, unavailable articles, and articles from legal documents reserved for evaluation are excluded. Section 4.7 describes the construction of these training examples.

3.6 Target Matching and Evaluation Metrics

Evaluation compares ranked articles with the relevance targets recorded for each query. An article-level target requires both the legal document identifier and the article identifier to match. A target with an empty article identifier or `toanvan` refers to the whole legal document, so any article from that document can match it. Each target is counted at most once. Retrieving several articles from the same document therefore does not increase the score for a single whole-document target. When a query has both types of target, the evaluator's target-checking order determines which unmatched target receives credit.

In the following formulas, Q is the set of evaluation queries, and R_q is the set of distinct recorded relevance targets for query q. The value $y_{q,r}$ is one if the article at rank r matches a previously unmatched target, and zero otherwise. The cutoff k specifies how many articles from the top of the ranking are evaluated.

Normalized discounted cumulative gain. Normalized discounted cumulative gain (NDCG) assigns more credit to relevant results at higher ranks and compares the observed ranking with an ideal ranking [18]. The thesis uses NDCG@10 as the primary metric, with

binary credit defined by the target-matching rules above:

$$\mathrm{NDCG@10}(q) = \frac{\sum_{r=1}^{10} y_{q,r} / \log_2(r+1)}{\sum_{r=1}^{\min(|R_q|,10)} 1/\log_2(r+1)}.$$

The denominator represents an ideal ranking in which up to ten targets are matched at the highest ranks. Queries without targets receive zero. Targets that cannot be matched to indexed articles remain in the denominator. The score therefore reflects both ranking quality and limitations in corpus coverage or identifier mapping.

Hit and recall. Hit@k is one if at least one target is matched within the first k results, and zero otherwise:

$$\mathrm{Hit@}k(q) = \mathbb{1}\left[\sum_{r=1}^{k} y_{q,r} > 0\right],$$

Here, $\mathbb{1}[\cdot]$ equals one when its condition is satisfied and zero otherwise. Hit is reported at $k = 1, 5, 10$. Recall measures the proportion of recorded targets that are retrieved [1]. Under the matching rules used here, it is calculated for queries with at least one target as

$$\mathrm{Recall@}k(q) = \frac{\sum_{r=1}^{k} y_{q,r}}{|R_q|}.$$

For example, matching one of two targets within the first k results gives Hit@k = 1 and Recall@k = 0.5. The metrics coincide for queries with one target. The baseline results report Hit, while the fine-tuning comparisons also report Recall@100 (Tables 5.11 and 5.12).

Mean reciprocal rank. Mean reciprocal rank (MRR) measures how early the first target is matched. For a query whose first matching article occurs at rank r_q^*, the reciprocal rank (RR) at cutoff ten is

$$\mathrm{RR@10}(q) = \begin{cases} 1/r_q^*, & \text{if a target is matched within the first ten results,} \\ 0, & \text{otherwise.} \end{cases}$$

MRR@10 is the mean of these values across queries:

$$\mathrm{MRR@10} = \frac{1}{|Q|} \sum_{q \in Q} \mathrm{RR@10}(q).$$

It considers only the first match, so additional matches do not change the score.

Mean average precision. Average precision (AP) combines precision values at ranks where relevant results are retrieved. Mean average precision (MAP) averages AP across

queries [1]. In this thesis, $P_q(r)$ is the proportion of the first r results that match distinct targets. The evaluator computes the metrics at cutoff ten as

$$P_q(r) = \frac{1}{r} \sum_{j=1}^{r} y_{q,j},$$
$$\text{AP@10}(q) = \frac{\sum_{r=1}^{10} y_{q,r} P_q(r)}{\max(1, \min(|R_q|, 10))},$$
$$\text{MAP@10} = \frac{1}{|Q|} \sum_{q \in Q} \text{AP@10}(q).$$

The AP@10 denominator is the number of recorded targets, capped at ten. The lower bound of one avoids division by zero and gives queries without targets a score of zero. This implementation differs from an AP definition that divides by the total number of targets without a cap.

Each aggregate metric gives equal weight to each query and includes queries with zero scores, unless the results explicitly restrict evaluation to queries with available targets. These metrics measure retrieval against the recorded annotations. They do not establish that every legally relevant article has been retrieved or that a generated legal answer is correct.

3.7 Chapter Summary

This chapter explains how BM25 and dense retrieval models score legal articles, how score fusion combines their rankings, and how fine-tuning updates the BGE-M3 embedding model. It also defines the target-matching rules and evaluation metrics used to compare retrieval performance. Chapter 4 describes the experimental setup and evaluation procedures.

Chapter 4

Research Methodology

This chapter describes the research design, datasets, and implementation used to evaluate Vietnamese legal article retrieval. It presents corpus preparation and BCA dataset construction before the retrieval pipeline, score fusion, and BGE-M3 fine-tuning procedures. It then specifies the evaluation design, statistical analysis, experimental environment, and data-use limitations.

4.1 Research Design and Retrieval Task

This thesis uses controlled retrieval experiments to compare lexical retrieval, dense retrieval, and their combination for Vietnamese legal questions. The system receives a question, scores the indexed legal articles, and returns a ranked list. Each article has a legal document identifier and an article identifier. The primary outcome is normalized discounted cumulative gain at rank ten (NDCG@10); the matching rules and evaluation metrics are defined in Section 3.6.

BM25 provides the baseline for retrieval effectiveness. Comparing BM25+BGE-M3 with BM25 assesses the benefit of combining lexical and dense retrieval. Comparing BM25+BGE-M3+E5 with BM25+BGE-M3 assesses the additional benefit of multilingual-E5-base. Dataset and corpus conditions provide separate settings for these comparisons, with the same query population retained within each comparison of corpus versions. The evaluation also distinguishes articles absent from the corpus from errors in mapping target identifiers, since both can affect measured performance.

The fine-tuning experiments compare pretrained and fine-tuned BGE-M3 in standalone dense retrieval and in the hybrid configurations. One experiment selects the model using hybrid retrieval performance on development data; the other uses standalone dense retrieval performance. The hybrid comparisons keep BM25, E5, and the combination weights fixed while replacing BGE-M3 with its fine-tuned version. This design tests whether improvements in standalone retrieval are retained after score fusion. Section 4.7 describes the training procedures, and Section 4.8 specifies the evaluation conditions and their limitations.

Relevance labels support training, configuration selection, and evaluation. At retrieval time, the system uses only the question text and the indexed corpus. Published answers

and recorded relevance targets are not retrieval inputs. The experiments therefore measure recovery of annotated legal articles rather than the correctness of a generated legal answer.

4.2 Datasets and Corpus Preparation

The overall retrieval benchmark uses the supplied reference article text for ALQAC and Zalo. All methods within a dataset use the same ordered queries, corpus, and relevance targets. BCA uses the separately constructed corpus described in Section 4.3. The completed Zalo fine-tuning experiments use their recorded training and evaluation corpora, described below and in Section 4.7; their results are separate from the reference-text benchmark.

Table 4.1 summarizes the three baseline datasets. Queries that share legal references are grouped for selection and evaluation, as explained in Section 4.8.

Table 4.1: Baseline datasets.

Dataset	**Queries**	**Articles**	**Evaluation design**
ALQAC	811	3,363	Grouped cross-validation using reference article text
Zalo	3,196	61,425	Grouped cross-validation using reference article text
BCA	546	14,789	Separate evaluation of the largest linked group and grouped cross-validation over smaller groups

4.2.1 ALQAC and Zalo Datasets

ALQAC and Zalo provide Vietnamese legal questions with annotations identifying the relevant legal articles. The questions serve as retrieval queries, and the annotated references provide the targets used to evaluate the ranked results.

For ALQAC, the baseline evaluation combines the supplied training and private-test questions, giving 811 queries. Its reference corpus contains 3,363 articles.

For Zalo, the baseline evaluation uses 3,196 queries and a reference corpus of 61,425 articles. Each query retains its supplied relevance targets.

For both datasets, questions linked by shared legal references are kept in the same group. Fusion weights are selected using other groups before evaluation on the excluded group, as described in Section 4.8. All baseline methods within each dataset use the supplied

reference corpus and annotations. The following subsection distinguishes this corpus from the preparation used in the completed fine-tuning experiments.

4.2.2 Article Extraction and Identifier Alignment

Each reference article record contains a legal document identifier, an article identifier, and article text. The overall ALQAC and Zalo benchmark uses these supplied records without comparing them with an extracted-text version.

The completed Zalo fine-tuning experiments use article text extracted from legal documents collected in raw HTML format. Preparation follows the reference article order. Extracted text replaces the reference text only if both document and article identifiers match and the cleaned text is nonempty. Otherwise, the reference text is retained. This preparation preserves 61,425 article identifiers and is documented here as the corpus actually used in those experiments; their scores are not relabeled as reference-text benchmark results.

Identifier normalization is separate from text preprocessing. Within each experiment, all retrievers use the same article order so that scores at the same position refer to the same article. Incorrect identifier alignment can prevent a relevant retrieved article from receiving evaluation credit even when its text is present.

4.3 BCA Dataset Construction

The BCA dataset is constructed from public questions and published answers on the Bộ Công an portal. Each question becomes a retrieval query, and legal references extracted from its published answer become the recorded relevance targets. Construction consists of collecting the source records, cleaning their text, extracting legal references, and mapping those references to corpus identifiers. The answer text is used to construct targets and is not supplied to the retriever.

4.3.1 Motivation and Research Value

The first purpose is to extend evaluation to questions submitted through a public-service channel. Competition benchmarks provide defined tasks and evaluation conditions; for example, ALQAC evaluates legal retrieval and question answering using a manually annotated dataset [8]. BCA adds questions concerning residence, identity documents, fire prevention, traffic penalties, and immigration. Citizens may describe their circumstances in everyday language rather than use the terms found in legislation. This motivates examining retrieval on public-service questions alongside the ALQAC and Zalo benchmark comparisons.

The second purpose is to construct a research resource beyond the supplied benchmarks. The collection and reference-extraction pipeline converts published question-answer records into queries and traceable relevance targets. It supports evaluation of BM25, BGE-M3, E5, and their combinations on a separately collected set of questions. The resulting analysis examines performance across collections, including differences in target availability and shared legal references. The resource contribution is the constructed collection and its documented processing procedure.

4.3.2 Data Sources and Record Selection

The collection script requests successive pages of question listings through the portal's application programming interface, then retrieves each question and its published answer. Each record is assigned an identifier derived from the source question identifier. During a repeated collection, records with existing identifiers are skipped to prevent duplication.

Acquisition requires separate listing and detail requests: the listing identifies available questions, while each detail response supplies the question and answer fields. Failed listing requests can interrupt discovery, and failed detail requests can leave individual records uncollected.

The saved collection contains 652 records with distinct question identifiers. Of these, 546 are retained for retrieval evaluation because they contain at least one extracted legal document reference.

4.3.3 Text Cleaning and Reference Extraction

Preprocessing covers the collected question-answer records, extracted legal articles, and inputs to each retriever. Legal references are then derived from the published answers to construct evaluation targets. The question text remains the retrieval query throughout this process.

Text cleaning and retriever inputs. The preprocessing steps are organized by their role:

1. **Cleaning question-answer records.** The response fields `content` and `answer` contain the question and published answer. Regular expressions replace HTML tags with spaces, remove leading and trailing whitespace, and reduce repeated spaces, tabs, and line breaks to a single space. Unicode Normalization Form C (NFC) standardizes equivalent character sequences in Vietnamese text. The cleaned question is used as the retrieval query, while the cleaned answer is used for reference extraction.

2. **Extracting legal articles.** Legal HTML is converted to text lines using BeautifulSoup. Article boundaries are identified from lines beginning with "Điều" and its alternative spelling, followed by an article number and an optional letter suffix. The leading article marker and number are removed where present.

3. **Preparing BM25 tokens.** Question and article text is converted to lowercase. Punctuation and other characters outside the word-character and whitespace classes are replaced with spaces, and the text is split on whitespace. Vietnamese letters with diacritics are retained. This procedure uses no external Vietnamese word-segmentation library.

4. **Formatting dense inputs.** BGE-M3 receives question and article text without additional prefixes. Multilingual-E5-base receives the prefix `query:` for questions and `passage:` for articles. Both models use a maximum input length of 512 tokens in the baseline experiments. Section 4.5 describes the retrieval models, and Section 4.7 specifies the separate fine-tuning configurations.

Legal reference extraction. The thesis processes the published answer using the following steps:

1. **Identifying articles and documents.** The answer is divided into spans using punctuation and line boundaries. Within each span, the extractor identifies article numbers, including forms such as "Điều 5", "Điều 12a", and "Điều 32". It also detects document terms such as "Luật", "Bộ luật", "Nghị định", and "Thông tư".

2. **Filtering ambiguous phrases.** Keywords such as "quy định", "quyết định", and "nghị quyết" require a following number to be treated as document references. Ordinary expressions such as "luật sư" and "luật chơi" are excluded. The rules also filter references marked as drafts and remove trailing phrases, incomplete parentheses, and other text outside the detected document name. Document-specific exceptions handle cases not covered by the general rules.

3. **Linking articles to documents.** The extractor first attempts to link an article mention to the following document mention when the intervening words are permitted connectors. If that association fails, it checks the preceding document using a separate connector-word set. This handles both "Điều 9 Nghị định 79/2014/NĐ-CP" and "Nghị định số 79/2014/NĐ-CP tại Điều 9, Điều 15". A document cited without a specific article is recorded with an empty article identifier, representing a whole-document target.

4. **Removing duplicate references.** Repeated document and article pairs are removed. When two extracted names refer to the same article and one name is

contained in a longer version, the more complete name is retained. For example, "Luật Giao thông đường bộ năm 2008" is retained over the shorter "Luật Giao thông đường bộ" when both have been extracted for the same article.

The extracted references are saved with the cleaned question-answer record. Section 4.3.4 describes their mapping to corpus identifiers and the checks for target availability. Evaluation uses the matching rules in Section 3.6: a whole-document target can match any indexed article from that document, and each recorded target receives credit at most once. Figure 4.1 summarizes the construction process and its connection to corpus coverage checks.

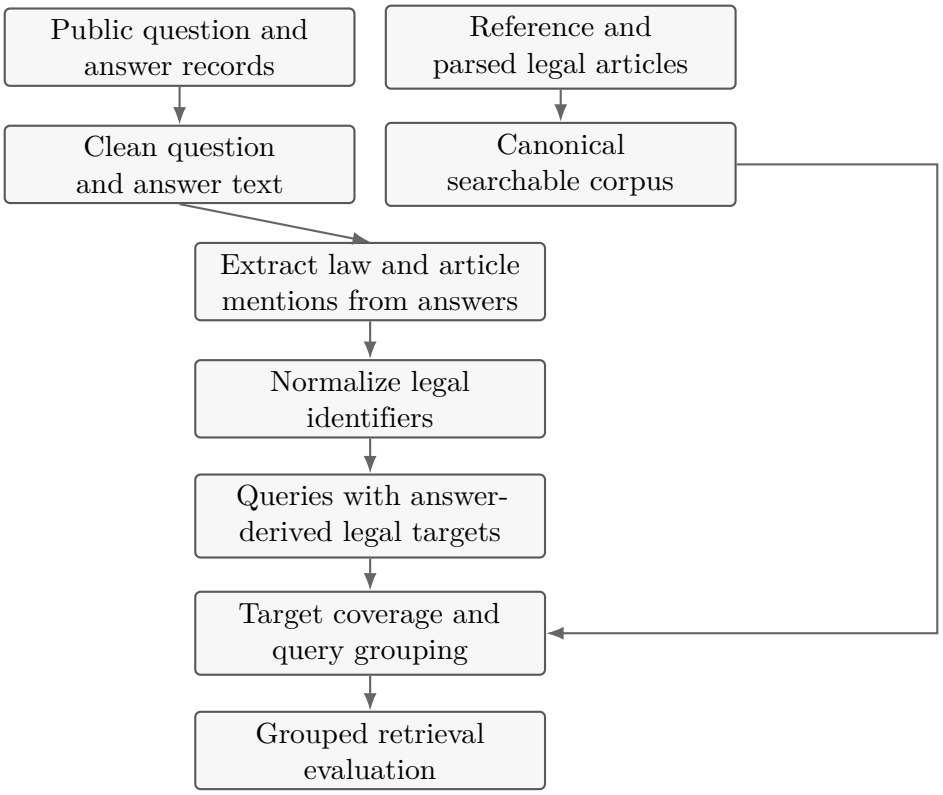


Figure 4.1: Construction of the BCA retrieval collection. Published answers supply reference evidence; only question text is used as the retrieval query.

4.3.4 Target Mapping and Corpus Coverage

Extracted legal references are mapped to corpus identifiers to construct evaluation targets. A correction table associates extracted document names with document identifiers. For example, "Nghị định số 100/2019/NĐ-CP" is mapped to `100/2019/nd-cp`. Leading and trailing whitespace is removed from document and article identifiers, and both are converted to lowercase. Each target consists of a document identifier and an article identifier. The article identifier is left empty when the published answer cites a document

without specifying an article. These normalization steps standardize identifiers but do not verify that the extracted reference has been mapped to the correct legal document.

The reference corpus is extended with legal documents collected from public sources in raw HTML format. Article text is extracted from these documents and added to the corpus, while the raw HTML and source mappings are saved. Duplicate article identifiers within each document and articles with empty text are excluded. The resulting searchable corpus contains 14,789 articles.

References that remain unresolved may be assigned the placeholder `INTERNAL_DOCUMENT`. Entries with this placeholder are excluded from corpus additions, but their questions and recorded targets remain in the evaluation set. The placeholder indicates an unresolved mapping, it does not establish that the referenced document is internal or access-restricted.

Target coverage is checked against the indexed article identifiers using the matching rules in Section 3.6. An article-level target requires a match on both the document and article identifiers. A whole-document target requires at least one indexed article with the same document identifier. A question has full coverage when all its recorded targets have a match, partial coverage when only some have a match, and no coverage when none has a match. These categories measure coverage of the recorded targets under the matching rules, rather than completeness of the corpus for answering the legal question.

A target without a match can result from missing corpus content or an unresolved or incorrect mapping. If the referenced article is absent from the corpus, no ranking method can retrieve it. If its text is present but the target is mapped incorrectly, retrieval may recover the relevant article without receiving evaluation credit. For a correctly mapped target present in the corpus, failure to retrieve a match within the evaluation cutoff concerns ranking performance. Therefore, coverage checks alone cannot establish whether a retrieval failure is caused by missing content, incorrect mapping, or ranking.

4.3.5 Example of Query and Target Construction

Table 4.2 illustrates how a question about fire-prevention equipment in apartment buildings becomes a retrieval query with recorded relevance targets. The published answer cites Nghị định số 79/2014/NĐ-CP. Reference extraction produces one whole-document target and two article-level targets, for Articles 9 and 15. The document identifier is normalized according to Section 4.3.4, while the question text supplies the retrieval query.

Table 4.2: Worked BCA question-to-target transformation.

Stage	**Representation**	**Interpretation**
Stored query	"Bộ Công an cho tôi hỏi có quy định bắt buộc nào về các trang thiết bị, dụng cụ phòng cháy chữa cháy đối với nhà chung cư không?"	Question text supplies the retrieval input
Extracted answer references	Nghị định số 79/2014/NĐ-CP, with article identifiers empty, 9, and 15	The saved extraction contains one whole-document reference and two article references
Normalized document identifier	`79/2014/nd-cp`	The mapped identifier is converted to lowercase
Evaluation targets	`(79/2014/nd-cp, empty)` `(79/2014/nd-cp, 9)` `(79/2014/nd-cp, 15)`	Each pair contains a document identifier and an article identifier; `empty` denotes an empty article identifier
Retrieval inputs	Question text as the query; indexed article text as the corpus	The published answer is used only to construct targets, which are used for evaluation

The empty article identifier allows any indexed article from this decree to match the whole-document target. The other two targets require the corresponding article identifiers. Under the rules in Section 3.6, each retrieved article receives credit for at most one previously unmatched target. When it could match both a whole-document target and an article-level target, the evaluator's target-checking order determines which target receives credit. Retrieving several articles from the decree does not repeatedly earn credit for the whole-document target.

4.4 Overview of Retrieval Experiments

The retrieval experiments use Vietnamese questions as queries and legal article text as the searchable corpus. BM25 computes lexical scores, while BGE-M3 and multilingual-E5-base, referred to as E5, compute dense similarity scores. Each retriever scores every article in the same corpus order, ensuring that scores combined across retrievers refer to the same article. The experiments evaluate BM25, BGE-M3, and E5 individually, as well as the hybrid configurations BM25+BGE-M3 and BM25+BGE-M3+E5.

For hybrid retrieval, each retriever's scores are normalized separately for each query over the full corpus. The normalized scores are combined using a weighted sum, and articles

are ranked by the combined score. Fusion weights are selected on the weight-selection queries and fixed before ranking the separate evaluation queries. The system returns the ten highest-ranked articles with their document and article identifiers. The returned identifiers are compared with the recorded relevance targets to compute NDCG@10 and the other evaluation metrics. Section 4.6 specifies score normalization and fusion, and Section 4.8 describes query grouping and the allocation of queries to weight selection and evaluation. Figure 4.2 summarizes this procedure with pretrained dense models.

The fine-tuning comparisons replace pretrained BGE-M3 with a fine-tuned model while keeping BM25, E5, and the corresponding fusion weights fixed. Standalone dense retrieval is evaluated separately to examine whether changes in BGE-M3's retrieval performance are also observed in the hybrid configurations. Section 4.7 describes the training and model selection procedures. The evaluated system produces article rankings; reranking and answer generation are outside its scope.

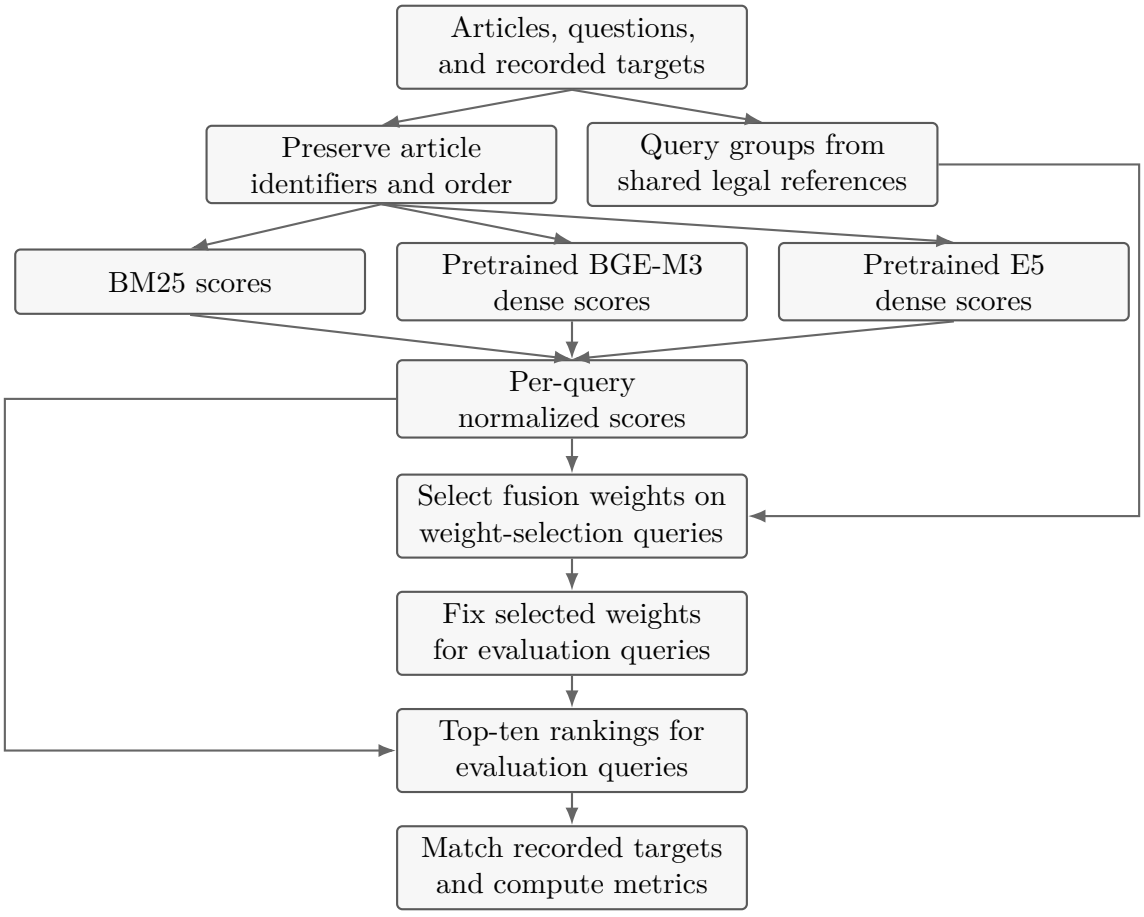


Figure 4.2: Hybrid scoring and evaluation pipeline with pretrained dense models. Scores are computed over the full corpus. Recorded targets support query grouping; targets for the weight-selection queries guide fusion-weight selection, and targets for the separate evaluation queries are used to compute metrics. Retrievers receive question and article text.

4.5 Retrieval Models and Implementation

The experiments use BM25 for lexical retrieval and the dense representations of BGE-M3 and multilingual-E5-base for semantic retrieval. All models score the same ordered articles within each corpus version. Preserving article order allows their scores to be combined without changing the associated legal document and article identifiers.

4.5.1 BM25

BM25 scores tokenized question and article text. Preprocessing converts text to lowercase, replaces non-word punctuation with token separators, and splits the resulting text on whitespace. This preprocessing remains fixed in the reported comparisons. The experiments therefore evaluate this BM25 configuration rather than alternative Vietnamese tokenization methods.

4.5.2 BGE-M3 and Multilingual-E5-base

BGE-M3 is the primary dense retriever. Multilingual-E5-base, referred to as E5, provides an additional dense score for the BM25+BGE-M3+E5 configuration. The experiments use BGE-M3's dense representation; its sparse and multi-vector retrieval methods are not evaluated.

Each encoder maps the question and each article to vectors. The implementation divides each vector by its Euclidean norm, with a minimum denominator of 10^{-10}, and computes the dot product of the normalized vectors. For the query vector $\mathbf{u}_q$ and article vector $\mathbf{v}_d$ before normalization, the resulting score is

$$s_{\text{dense}}(q, d) = \frac{\mathbf{u}_q^\top \mathbf{v}_d}{\max(\|\mathbf{u}_q\|_2, 10^{-10}) \max(\|\mathbf{v}_d\|_2, 10^{-10})}.$$

Here, q is the question, d is an indexed article, and $\|\cdot\|_2$ denotes the Euclidean norm. For vectors whose norms exceed the numerical threshold, this calculation is cosine similarity. The threshold prevents division by zero.

Table 4.3 summarizes the pretrained dense models and the baseline input settings. Both encoders use a maximum input length of 512 tokens in the baseline experiments. This experimental limit is distinct from BGE-M3's supported maximum of 8,192 tokens described in Chapter 3. The fine-tuning experiment selected by hybrid retrieval performance uses maximum lengths of 256 tokens for queries and 512 tokens for passages, as specified in Section 4.7.

Table 4.3: Baseline dense-model settings.

Property	**BGE-M3**	**Multilingual-E5-base**
Model family	BGE-M3 multilingual embedding model	E5 multilingual text-embedding model
Embedding dimension	1,024	768
Experimental role	Primary pretrained dense component	Additional pretrained dense component
Baseline maximum input length	512 tokens	512 tokens
Fine-tuning status	Fine-tuned in the experiments in Section 4.7	Pretrained parameters remain fixed

The evaluation scores every indexed article rather than restricting retrieval to candidates from an approximate nearest-neighbor index. Thus, an available target that falls outside the first ten results is excluded by the resulting ranking rather than by a separate candidate filter. This exhaustive scoring design supports comparisons of ranking performance; it does not establish the system's latency in deployment.

4.6 Score Normalization and Fusion

BM25 and dense retrieval produce scores on different scales. Before combining them, the baseline applies Min-Max normalization separately to each retriever's scores for each query. For retriever r, raw score $s_r(q, d)$, and full indexed corpus C, the normalized score is

$$\widetilde{s}_r(q, d) = \frac{s_r(q, d) - \min_{d' \in C} s_r(q, d')}{\max\left(\max_{d' \in C} s_r(q, d') - \min_{d' \in C} s_r(q, d'), \epsilon\right)}.$$

Here, ϵ is a small positive constant that prevents division by zero. If a retriever assigns the same score to every article for a query, all its normalized scores are zero. Normalization uses the full corpus rather than a shortlist of retrieved candidates, so every indexed article can enter the final ranking.

The BM25+BGE-M3 score is

$$S_{bg}(q, d) = w_b \widetilde{s}_b(q, d) + (1 - w_b)\widetilde{s}_g(q, d),$$

where b denotes BM25, g denotes BGE-M3, and $0 \leq w_b \leq 1$. Adding E5 gives

$$S_{bge}(q,d) = w_b\widetilde{s}_b(q,d) + w_g\widetilde{s}_g(q,d) + w_e\widetilde{s}_e(q,d),$$
$$w_b + w_g + w_e = 1,$$

where e denotes E5 and all weights are nonnegative. The articles are ranked by the combined score in decreasing order.

The baseline selects weights from a grid with increments of 0.1. BM25+BGE-M3 has 11 candidate settings, with $w_b \in \{0, 0.1, \ldots, 1\}$. BM25+BGE-M3+E5 has 66 candidate settings whose weights sum to one. This grid includes zero-weight settings, including $w_e = 0$, so it can reproduce every BM25+BGE-M3 setting in the baseline grid. A higher score on the weight-selection queries therefore does not establish that E5 improves retrieval on separate evaluation queries.

For each baseline split, the weights with the highest mean NDCG@10 on the weight-selection queries are fixed before evaluation on the other queries. Section 4.8 describes the grouping and fold procedure. Algorithm 4.1 shows how the selected configuration produces a ranking for one query.

Algorithm 4.1 Article ranking with fixed fusion weights

Require: Query q, corpus C, retriever scores, fixed fusion weights w
Ensure: Ten highest-ranked articles
1: Normalize each retriever's scores over the full corpus for q
2: **for** each article $d \in C$ **do**
3: combined$[d] \leftarrow \sum_r w_r$ normalized_score$[r, d]$
4: **end for**
5: **return** the ten articles with highest combined score

The fine-tuning comparisons use weights selected with pretrained BGE-M3 and keep them fixed when replacing it with the fine-tuned model. BM25 and E5 also remain unchanged. These comparisons measure the effect of replacing BGE-M3 within the specified hybrid configuration. The 0.1 grid described above applies to baseline selection. The fine-tuning experiments use separate selection procedures, described in Section 4.7.

4.7 BGE-M3 Fine-Tuning

The two fine-tuning experiments differ in how they select BGE-M3. The first selects the model using hybrid retrieval performance on development queries, followed by evaluation on a separate test set. The second selects it using standalone dense retrieval performance, then evaluates the selected model both on its own and in BM25+BGE-M3 and BM25+BGE-M3+E5. Within each hybrid comparison, fine-tuned BGE-M3 replaces pretrained BGE-M3 while BM25, E5, and the selected fusion weights remain fixed. These comparisons test

whether fine-tuning improves standalone retrieval and whether any improvement is retained in the hybrids under fixed fusion weights. The experiments use different training procedures and are therefore described separately. Section 5.3.2 reports the results and evaluation limitations.

4.7.1 Model Selection Based on Hybrid Retrieval

Training Data and Negative Examples

This experiment uses Zalo question and article pairs. For each eligible query, the training builder selects a known relevant article available in both clean and parsed views, creating paired training examples from the two corpus representations. Negative passages are mined from lexical and dense rankings after excluding known positives, duplicates, unavailable documents, and laws reserved for development and test evaluation.

The negative-selection procedure considers up to 256 candidates from each of BM25 and pretrained BGE-M3 on the parsed corpus. It alternates between the filtered lists until seven distinct negatives have been selected. These rankings and negative groups are fixed before training. All explicit negatives use parsed article text, including groups whose positive uses clean text.

Table 4.4: Training groups and protected folds in the completed hybrid fine-tuning study.

Stage	Train folds	Queries	A/B groups	Protected folds	Protected laws	Law overlap
Pilot	2, 3, 4	1,917	3,834	0, 1	258	0
Final training	1, 2, 3, 4	2,556	5,112	0	110	0

Each eligible query produces two training groups using the clean and parsed text of the same selected relevant article. The group counts therefore do not represent independent questions. The zero-overlap checks confirm that training groups contain no article from the protected law identifiers; they do not establish separation of legal concepts or similar language. Explicit-negative filtering also does not guarantee that passages from other groups in the same batch are nonrelevant to a query.

Training Objective and Configuration

This experiment trains BGE-M3 with a single-positive in-batch contrastive objective. Each query group contains one selected positive and seven explicit negatives. With batch size B, group size $G = 8$, positive passage index $y_i = iG$, normalized query vector $\mathbf{q}_i$, normalized

passage vector $\mathbf{p}_j$, and temperature $\tau = 0.02$, the implemented objective is

$$\mathcal{L}_{\text{batch}} = -\frac{1}{B}\sum_{i=0}^{B-1}\log\frac{\exp(\mathbf{q}_i^\top \mathbf{p}_{y_i}/\tau)}{\sum_{j=0}^{BG-1}\exp(\mathbf{q}_i^\top \mathbf{p}_j/\tau)}.$$

Table 4.5: Locked configuration for the completed final-training runs.

Parameter	**Verified value**
Base encoder	BGE-M3
Trainable component	Full dense encoder
Epochs	2
Learning rate	2.0000×10^{-5}
Contrastive query batch	4
Gradient accumulation	8
Optimizer effective batch	32 query groups
Passage group	1 selected positive and 7 explicit negatives
Query / passage maximum length	256 / 512 tokens
Temperature	0.0200
Pooling / normalization	First-token hidden state / L2 normalization
Seeds	42, 43, 44
Frozen E5 component	Multilingual-E5-base, unchanged throughout fine-tuning

The numerator rewards the selected positive article, while the denominator includes all passages in the forward-pass batch. With four query groups and eight passages per group, this denominator contains 32 passage positions. Gradient accumulation across eight forward passes yields an effective optimizer batch of 32 query groups, but does not combine their passages into one contrastive denominator.

Model Selection and Evaluation

Pilot training uses folds 2, 3, and 4. Fold 1 selects the learning rate, epoch, and forward-pass query batch using hybrid NDCG@10. NDCG@10 for BM25+BGE-M3 is the primary selection criterion, while BM25+BGE-M3+E5 is checked separately. The development rule requires an improvement of at least 0.01 for BM25+BGE-M3 and a decrease of no more than 0.01 for BM25+BGE-M3+E5. After the configuration is fixed, final models are trained again from pretrained BGE-M3 on folds 1, 2, 3, and 4 with seeds 42, 43, and 44.

Fold 0 contains 640 queries reserved for evaluation on a separate test set. The corrected comparison selects combination weights on folds 1, 2, 3, and 4 using pretrained BGE-M3 and keeps them fixed when replacing it with fine-tuned BGE-M3. A preliminary analysis had already examined fold 0 before a document-identity error was corrected. The corrected results are therefore interpreted as post-hoc evidence, despite excluding fold 0 from training and model selection.

Figure 4.3 summarizes the selection and evaluation sequence.

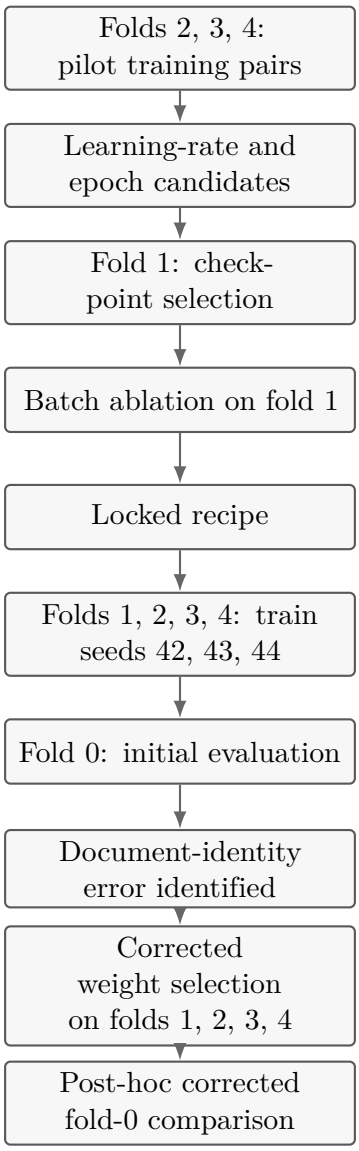


Figure 4.3: Training and evaluation evidence flow.

4.7.2 Model Selection Based on Standalone Dense Retrieval

Training Data and Configuration

This experiment selects BGE-M3 using standalone dense retrieval before testing it in BM25+BGE-M3 and BM25+BGE-M3+E5. It uses a fixed corpus of 61,425 parsed articles, with reference text retained where parsed text is unavailable. The official training partition contains 2,556 queries. Removing one near-duplicate of an official test query leaves 2,555 eligible training groups. The final evaluation uses 640 official test queries.

Each group includes five explicit negatives. The training procedure uses a mask to identify all known positive articles within the batch candidate set. This differs from the selected-positive objective in Section 4.7.1. The encoder uses the hidden state at the first token position, L2 normalization, and query and passage limits of 256 and 512 tokens, respectively. The specified temperature is 0.05, with bfloat16 precision, a cosine learning-rate schedule, and a warmup ratio of 0.05. The procedure requires at least 32 queries in a forward-pass contrastive batch, with 64 preferred; this requirement cannot be met by gradient accumulation alone. The available description does not identify which of these batch sizes was used in every final run.

Training Objective

For logits $z_{ij} = s(q_i, d_j)/\tau$, positive candidate positions P_i, candidate positions C_i, and batch size B, the multi-positive loss is

$$\mathcal{L} = -\frac{1}{B}\sum_{i=1}^{B}\left[\log\sum_{j\in P_i}\exp(z_{ij}) - \log\sum_{j\in C_i}\exp(z_{ij})\right].$$

Here $s(q_i, d_j)$ is the similarity between query i and candidate article j, and $\tau = 0.05$ controls the scale of the logits. The numerator includes every known positive present in the candidate set rather than one selected positive. This use of recorded labels does not establish that every unannotated candidate is legally irrelevant. The documented checks concern article identity, labels, and data provenance; they do not constitute expert review of every negative example.

Model Selection and Hybrid Evaluation

Screening evaluates 36 checkpoints across three learning rates, four development folds, and training steps 30, 60, and 90. A training-data stage passes if its mean NDCG@10 gain is at least 0.01, at least three folds improve NDCG@10, and mean Recall@100 decreases by no more than 0.01. These criteria use standalone dense retrieval. The Zalo-only stage is the first stage to pass, so subsequent data expansions are skipped. Screening selects a

learning rate of 2×10^{-5} and step 60. Final models are trained using seeds 42, 43, and 44, with each checkpoint taken at step 60 and checked after reloading.

For the hybrid comparisons, combination weights are selected on official training queries using pretrained BGE-M3, then fixed when fine-tuned BGE-M3 replaces it. BM25+BGE-M3 uses BM25 and BGE-M3 weights of 0.20 and 0.80. BM25+BGE-M3+E5 uses BM25, BGE-M3, and E5 weights of 0.15, 0.50, and 0.35. Dense and hybrid comparisons use the same 640 test queries. A gain of at least 0.01 NDCG@10 defines the practical improvement threshold, with the Recall@100 condition checked separately. This threshold is a decision rule rather than a statistical significance test.

The official test queries had been examined before this experiment. Excluding them from final training and model selection does not remove that prior exposure, so the results are interpreted as post-hoc evidence. They assess query retrieval over a fixed corpus and do not establish transfer to new legal documents or a later time period. Chapter 5 reports standalone and hybrid outcomes separately.

4.8 Evaluation Design and Statistical Analysis

4.8.1 Query Grouping and Data Splits

Queries that cite the same legal document are linked into groups. These links are transitive: if one question cites documents shared with two other questions, all three belong to the same connected group. Keeping each group within one fold prevents a cited document from appearing on both sides of fusion-weight selection and evaluation. It does not eliminate similarity in legal concepts or wording across groups.

For the pretrained ALQAC and Zalo comparisons, fusion weights are selected on the other folds and evaluated on the excluded fold. Each query receives one out-of-fold result, meaning that its relevance targets are excluded from the weight selection used for its prediction. The encoders remain fixed during this procedure. The ALQAC query pool combines the original training and private-test gold files, so these results describe grouped evaluation over that combined pool rather than performance on the original private-test partition.

Algorithm 4.2 specifies the selection procedure. The overall score is calculated from all evaluation queries, giving each query equal weight. It is not an unweighted mean of fold scores when fold sizes differ. Because each fold can select different weights, the result evaluates the selection procedure rather than one fixed weight vector.

Algorithm 4.2 Grouped selection of fusion weights and out-of-fold evaluation

Require: Ordered queries Q, corpus D, aligned normalized score matrices, grouped folds, ordered candidate weight grid W
Ensure: Selected weights per fold and evaluation metrics for every query
for each fold f **do**
    Set T to the selection-query indices and V to the evaluation-query indices
    for each candidate weight vector $w \in W$, in recorded order **do**
        Rank D for every query in T using the weighted score sum
        Compute mean query NDCG@10
    end for
    Select the first candidate with the highest selection mean
    Rank all queries in V using the selected weights
    Store each evaluation query's metrics
end for
Aggregate evaluation-query rows across folds, giving each query equal weight

BCA requires a separate allocation because 484 of its 546 queries form one connected group. The remaining 62 queries belong to 49 smaller groups. The first comparison selects weights using the 62 queries and evaluates on the 484-query group. The smaller groups are then assigned to four evaluation folds containing 16, 16, 15, and 15 queries. For each of these folds, weights are selected on the remaining queries, including the largest group. Each question receives one evaluation result, but the selection populations differ substantially. The combined BCA score is therefore a descriptive summary, not a balanced five-fold estimate.

The fine-tuning experiments use the separate training, development, and test allocations described in Section 4.7. Their results are compared with pretrained models on the same test queries, not with the all-fold pretrained averages. Although test queries are excluded from the corresponding training and model selection, they had been examined in earlier analyses. Both fine-tuning comparisons are therefore interpreted as post-hoc evidence.

4.8.2 Evaluation Metrics and Comparisons

The target-matching rules and metric formulas are defined in Section 3.6. NDCG@10 is the primary metric. Hit@10 measures whether at least one target is recovered, MRR@10 measures the position of the first match, and MAP@10 summarizes precision at matching ranks. The standalone dense and corresponding hybrid fine-tuning comparisons also report Recall@100.

The comparison of BM25+BGE-M3 with BM25 measures improvement over the lexical baseline. The comparison of BM25+BGE-M3+E5 with BM25+BGE-M3 isolates the contribution of adding E5 under the specified selection procedure. Fine-tuning comparisons replace pretrained BGE-M3 with its fine-tuned version while holding BM25, E5, and the corresponding fusion weights fixed. Standalone dense performance is assessed separately from hybrid performance. The specified threshold for a practically useful hybrid

improvement is an absolute increase of 0.01 NDCG@10; this threshold is distinct from a statistical significance criterion.

BCA results are reported both for all retained queries and for queries with at least one target available in the corpus. The latter population still includes queries with only some targets available, and their original metric denominators are retained. This distinction separates the effect of target availability from ranking performance without treating incomplete annotations as complete relevance judgments.

4.8.3 Statistical Analysis

For ALQAC and Zalo, the recorded comparison between BM25+BGE-M3+E5 and BM25+BGE-M3 uses the Wilcoxon signed-rank test on paired query-level out-of-fold NDCG@10 and MRR@10 values. Its `zero_method=zsplit` setting splits the ranks of zero differences between the positive and negative rank sums. These stored tests are unadjusted for the multiple condition and metric comparisons. Queries within the same legal group can remain dependent despite grouped weight selection, so the query-level tests do not establish variation across independent legal topics.

For BCA's 49 smaller groups, the analysis first averages paired NDCG@10 differences within each group. It then applies 100,000 random sign flips to those group means and uses 10,000 bootstrap resamples of the group means to form a 95% interval. Each group has equal weight in these procedures. Their interval therefore concerns the mean across groups, whereas the reported query-level mean gives each question equal weight. These quantities differ when group sizes differ. Query resampling within the single largest group describes variation among its questions and cannot estimate variation across multiple such legal groups. The combined BCA aggregate remains descriptive.

For the corrected fine-tuning comparison selected by hybrid performance, fine-tuned NDCG@10 is first averaged across seeds 42, 43, and 44 separately for each query. The matched pretrained value is then subtracted. The analysis uses 2,000 seeded bootstrap resamples of 106 legal groups covering the 640 test queries. Each resample draws groups with replacement, includes all their query differences, and calculates the query mean. Percentiles of these means form the reported 95% interval.

That implementation also records twice the smaller proportion of bootstrap means on either side of zero, capped at one, followed by Holm adjustment across the two hybrid comparisons. This is a bootstrap tail-area quantity, not a Wilcoxon result. Because the resamples come from the observed differences rather than an explicitly generated null distribution, it is not treated as a calibrated hypothesis-test p-value. Unequal group sizes also affect the resampling distribution: one legal group contains 428 of the 640 queries.

Averaging within each query avoids counting three seed-specific predictions as three

independent questions. The resulting interval concerns the average of the three observed seed results; it does not estimate uncertainty over arbitrary future seeds or evaluate an ensemble of encoders. For both fine-tuning experiments, variation across the three seeds is reported separately from uncertainty over queries. An interval containing zero does not establish equivalence, and a numerical improvement does not by itself establish a statistically supported or practically useful gain.

4.9 Experimental Environment and Reproducibility

The recorded execution environments differ between the two fine-tuning experiments. Both report an NVIDIA L4 graphics processing unit and PyTorch `2.11.0+cu128`. The experiment using hybrid retrieval for model selection records Transformers `5.15.0`, NumPy `2.1.3`, and 16-bit floating-point (FP16) precision. The experiment using standalone dense retrieval for model selection records Transformers `5.16.1` and bfloat16 (BF16) precision. These details describe the recorded training runs rather than the local computer used to prepare the thesis. The final hybrid-selected run records do not specify processor, system memory, graphics memory, or driver details; preliminary environment checks cannot establish those details for the completed runs.

Each selected fine-tuning configuration is evaluated with training seeds 42, 43, and 44. Reproducing the comparison requires the same model revision and tokenizer, training examples, query and article ordering, data splits, checkpoint selection rule, and fusion coefficients. Appendix D records the hybrid-selected configuration, model revision, and checksums for its model files, training data, and training code. These records identify the inputs used in the experiment; access to the checkpoint files is also required to reproduce encoder outputs.

The saved score matrices and selected fusion coefficients support recalculation of the reported retrieval metrics without retraining. Their rows and columns must remain aligned with the recorded query and article identifiers. Recalculating metrics verifies the numerical results but does not establish that model training can be repeated exactly. Appendix F describes the reproduction procedure and its limits.

The experiments assess retrieval effectiveness. They do not provide controlled measurements of query latency, throughput, or deployment cost, so the recorded hardware specifications do not support conclusions about those outcomes.

4.10 Research Ethics and Data Use

The experiments use the ALQAC and Zalo collections and the BCA dataset described in Sections 4.2 and 4.3. BCA is derived from public questions and published answers on the Bộ Công an portal. Public access identifies the source of these records, but does not by itself establish permission for unrestricted reuse or redistribution. The construction records described in this thesis do not establish consent from the people who submitted the questions or document a completed review of personal information.

Public questions and answers may contain details that identify individuals. Any future release of BCA therefore requires a review of identifying information, source terms, and the permitted scope of redistribution. The example in Section 4.3 summarizes the query and target structure without reproducing personal details. Published answers are retained as evidence for reference extraction and are not supplied as query inputs during retrieval.

The evaluation measures recovery of recorded legal references. These references are extracted from published answers and may be incomplete or incorrectly mapped, as discussed in Section 4.3. Retrieval scores therefore do not establish the correctness of a legal answer or the current applicability of a retrieved provision. The experiments assess article ranking and do not evaluate a system for providing legal advice.

4.11 Chapter Summary

This chapter presents the research design, the datasets and corpus preparation, and the construction of BCA from public questions and published answers. It specifies lexical and dense retrieval, score normalization and fusion, and BGE-M3 fine-tuning with model selection based on either hybrid or standalone dense retrieval. The evaluation design explains query grouping, data splits, metric comparisons, and the scope of statistical analysis. The chapter also documents the experimental environment, reproducibility requirements, and data-use limitations; Chapter 5 reports the results.

Chapter 5

Experiments and Evaluation

This chapter describes the experimental conditions and benchmark evaluation procedure, then presents the results and analysis. The results cover the Bộ Công an (BCA) collection, BGE-M3 fine-tuning relative to matched pretrained baselines, and the overall retrieval benchmark, including the contributions of hybrid retrieval and multilingual-E5-base (E5).

5.1 Experimental Conditions

The overall benchmark uses the supplied ALQAC and Zalo reference corpora, while BCA uses its separately constructed corpus. The retrieval models are BM25, `BAAI/bge-m3`, and `intfloat/multilingual-e5-base`. Baseline dense inputs are limited to 512 tokens. Dataset sizes, preprocessing, and model settings are specified in Sections 4.2 and 4.5.

Table 5.1 summarizes the recorded hardware and software for the two fine-tuning experiments.

Table 5.1: Recorded fine-tuning hardware and software.

Component	**Hybrid-based selection**	**Standalone dense selection**
Graphics processing unit	NVIDIA L4	NVIDIA L4
PyTorch build	`2.11.0+cu128`	`2.11.0+cu128`
Transformers	`5.15.0`	`5.16.1`
NumPy	`2.1.3`	Not recorded
Numerical precision	16-bit floating point (FP16)	bfloat16 (BF16)
Training seeds	42, 43, 44	42, 43, 44

The experiment were ran on a single NVIDIA L4 GPU on Google Colab platform. The reason is that the L4 GPU is a powerful and cost-effective option for training and fine-tuning machine learning models. It provides sufficient computational resources to handle the training of large language models like BGE-M3 and E5, while also being more accessible and affordable compared to higher-end GPUs. Additionally, Google Colab offers a convenient environment for running experiments without the need for extensive local hardware setup, making it an ideal choice for this research.

5.2 Benchmark & Evaluation Procedure

The overall benchmark evaluates ALQAC and Zalo using their supplied reference article text. BCA uses the corpus constructed from collected legal documents. Each fine-tuning experiment uses its recorded corpus and compares pretrained and fine-tuned models on the same test queries. The datasets and corpus preparation are described in Sections 4.2 and 4.3.

The evaluation follows six steps:

1. **Prepare and align the inputs.** Prepare question text, article text, and relevance targets using the procedures in Chapter 4. Check document and article identifiers, and keep the article order consistent across retrievers. Recorded targets are used for selection and evaluation, not as retrieval inputs.

2. **Separate selection and evaluation queries.** For the pretrained benchmark, keep questions linked by shared legal references in the same group. For ALQAC and Zalo, select fusion weights on the other folds and evaluate the excluded fold. For the reported BCA comparison, use 62 queries to select weights and all 484 queries in the largest group for evaluation. Fine-tuning uses the training, development, and test allocations in Section 4.7.

3. **Compute article scores.** BM25, BGE-M3, and E5 score the same ordered corpus within each comparison. For hybrid retrieval, normalize each retriever's scores separately for each query over the full corpus, then combine them using a weighted sum, as defined in Section 4.6.

4. **Select the configuration.** Choose fusion weights using mean normalized discounted cumulative gain at rank ten (NDCG@10) on the selection queries. For fine-tuning, select the model using either hybrid or standalone dense performance on development data, according to the corresponding procedure. Fix the selected configuration before test evaluation.

5. **Evaluate the rankings.** Compare ranked article identifiers with the recorded relevance targets under the rules in Section 3.6. NDCG@10 is the primary metric. Hit@10 measures whether a target is retrieved within the first ten results, mean reciprocal rank at ten (MRR@10) measures how early the first match occurs, and mean average precision at ten (MAP@10) summarizes precision at matching ranks. The standalone dense and corresponding hybrid comparisons also report Recall@100 using the first 100 results.

6. **Compare matched results.** Compare BM25+BGE-M3 with BM25 to assess improvement over the lexical baseline. Compare BM25+BGE-M3+E5 with

BM25+BGE-M3 to assess the contribution of E5. Compare fine-tuned with pretrained BGE-M3 on the same test queries, assessing standalone and hybrid performance separately. Within each hybrid fine-tuning comparison, BM25, E5, and fusion weights remain fixed.

ALQAC and Zalo baseline results give each query equal weight and combine its out-of-fold prediction; they do not average fold scores equally when fold sizes differ. The reported BCA benchmark describes one group of 484 questions and does not establish variation across independent legal groups. The fine-tuning test sets had been examined in earlier analyses, so the comparisons remain post-hoc despite their exclusion from the corresponding training and model selection. Statistical analysis and the practical improvement threshold are specified in Section 4.8.

Values are displayed to four decimal places, with scientific notation for small nonzero values. Differences are calculated before rounding. Each fine-tuning experiment uses its own matched pretrained baseline; results from different selection procedures are not pooled.

5.3 Experimental Results & Analysis

The results are organized into BCA dataset construction and retrieval, BGE-M3 fine-tuning, the overall retrieval benchmark, and a summary of the findings.

5.3.1 BCA Dataset Results

Collected Records and Target Coverage

The collection contains 652 distinct public question-answer records. Of these, 546 contain at least one extracted legal document reference and are retained for retrieval evaluation. The remaining 106 records are excluded because no reference was extracted. These excluded records is still a law question but the answers did not contain any legal references. The indexed corpus contains 14,789 articles.

Table 5.2: BCA collection size and target coverage.

Quantity	**Count**
Collected question-answer records	652
Retained evaluation queries	546
Indexed legal articles	14,789
Retained queries with all targets matched	211
Retained queries with some targets matched	229
Retained queries with no target matched	106

In total, 440 retained queries have at least one target matching an indexed article. Full coverage comprises 211 queries, calculated as 440 minus the 229 partially covered queries. Coverage measures identifier matches rather than mapping accuracy. No independently adjudicated mapping-accuracy estimate is available for this collection.

Observed Problems in Public-Service Questions

The collection records two distinct obstacles to constructing retrieval targets. Reference extraction yields no target for the 106 excluded records. These records is still a law question but the answers did not contain any legal references. This is a common problem in public-service questions, where the answer may describe a legal requirement without citing the relevant decree or article. The 229 partially covered queries illustrate another difficulty: the recorded targets are not fully matched to the indexed articles. the answers despite comming from official sources, still have incomplete or incorrect references. The 229 partially covered queries illustrate another difficulty: the recorded targets are not fully matched to the indexed articles.

These observations explain the difficulty of crawling and indexing public-service questions. The answers may not contain any legal references, or the references may be incomplete or incorrect.

Retrieval Benchmark on BCA

Table 5.3 compares BM25, pretrained BGE-M3, and the two hybrid configurations on all 484 queries in the largest group linked by shared legal references. Fusion weights are selected on the remaining 62 BCA queries and fixed before evaluation, as specified in Section 4.8. The results cover the 484-query group, including questions with partial or no target coverage, rather than the full set of 546 retained questions.

Table 5.3: Retrieval results on 484 BCA queries.

Method	**NDCG@10**	**MRR@10**	**MAP@10**	**Hit@10**
BM25	0.1670	0.2469	0.1098	0.4545
BGE-M3	0.2153	0.3143	0.1453	0.5455
BM25+BGE-M3	0.2275	**0.3414**	**0.1573**	0.5517
BM25+BGE-M3+E5	**0.2295**	0.3413	0.1560	**0.5702**

Pretrained BGE-M3 scores higher than BM25 on all four metrics, and both hybrids score higher than either individual retriever reported in the table. BM25+BGE-M3+E5 has the highest NDCG@10, the primary metric, and the highest Hit@10. BM25+BGE-M3 has the highest MRR@10 and MAP@10. Adding E5 therefore increases the observed NDCG@10

and Hit@10 but reduces the other two metrics; it does not improve every aspect of the ranking.

From this result we can say that hybrid retrieval improves the overall retrieval performance on the BCA dataset, with the combination of BM25, BGE-M3, and E5 yielding the best NDCG@10 and Hit@10 scores.

Another conclusion is, due to the nature of the BCA dataset, which contains public-service questions with everyday language, the result is sub 30% NDCG@10, which is lower than the other datasets. This indicates that the BCA dataset is more challenging for retrieval models, likely due to the informal language and incomplete or incorrect references in the questions and answers.

5.3.2 BGE-M3 Fine-Tuning Results

This section reports two fine-tuning experiments. The first selects BGE-M3 using hybrid retrieval performance. The second selects it using BGE-M3's standalone dense retrieval performance, then evaluates the selected model in the hybrids. The goal is to determine whether fine-tuning improves BGE-M3's retrieval performance, either as a standalone dense retriever or in hybrid configurations.

Selected Fine-Tuning Configurations

Table 5.4 lists the selected settings. The experiment run on the same dataset and iterate thought parameters to find the best configuration. The selected settings are based on the best performance of NDCG@10 on the development set. The selected settings are used for the final evaluation on the test set.

Table 5.4: Selected BGE-M3 fine-tuning parameters.

Setting	**Hybrid-based selection**	**Standalone dense selection**
Learning rate	2×10^{-5}	2×10^{-5}
Selected training duration	2 epochs	Checkpoint at step 60
Queries per forward pass	4	Not recorded
Gradient-accumulation steps	8	Not recorded
Explicit negative articles per query	7	5
Query / passage token limit	256 / 512	256 / 512
Contrastive temperature	0.02	0.05

Fine-Tuning for Hybrid Retrieval

Selecting the settings. This experiment select the best learning rate and number of epochs in a hybrid retrieval performance setup. It evaluates 18 configurations across three

learning rates and six epoch counts.

Table 5.5: Learning-rate and epoch selection on Zalo.

Learning rate	**Epoch**	**BM25+BGE-M3**	**BM25+BGE-M3+ E5**
5.0000×10^{-6}	1	0.7470	0.7685
5.0000×10^{-6}	2	0.7603	0.7785
5.0000×10^{-6}	3	0.7647	0.7815
1.0000×10^{-5}	1	0.7509	0.7816
1.0000×10^{-5}	2	0.7658	0.7873
1.0000×10^{-5}	3	0.7628	0.7897
2.0000×10^{-5}	1	0.7440	0.7757
2.0000×10^{-5}	2	**0.7691**	0.7923
2.0000×10^{-5}	3	0.7689	**0.7958**

The selected learning rate is 2×10^{-5}, with two epochs. This setting has the highest BM25+BGE-M3 NDCG@10 among the candidates. sec:finetuning-hybrid-selection.

Table 5.6: Query-batch comparison on Zalo development data.

Batch / steps	**BM25+BGE-M3**	**Δ**	**BM25+BGE-M3+E5**	**Δ**
Pretrained baseline	0.7172	N/A	0.7245	N/A
1 / 32	0.7725	+0.0553	0.7869	+0.0624
2 / 16	0.7693	+0.0521	0.7924	+0.0679
4 / 8	**0.7781**	**+0.0609**	**0.7934**	**+0.0689**

All three query-batch configurations increase development NDCG@10 for both hybrids relative to their pretrained baselines. The selected four-query batch has the highest score for each hybrid among these configurations.

Table 5.7: BM25+BGE-M3 before and after fine-tuning on Zalo fold 0.

BGE-M3 state	**Seed**	**NDCG**	**Δ**	**MRR**	**MAP**	**Hit**
Pretrained	Fixed	**0.7516**	N/A	**0.7015**	**0.6985**	**0.9141**
Fine-tuned	42	0.7005	**−0.0511**	0.6470	0.6451	0.8750
Fine-tuned	43	0.7005	**−0.0511**	0.6490	0.6467	0.8688
Fine-tuned	44	0.6977	−0.0539	0.6398	0.6374	0.8859
Fine-tuned	Mean	0.6995	−0.0520	0.6453	0.6430	0.8766

All three fine-tuned BM25+BGE-M3 runs have lower NDCG@10 than their matched pretrained baseline. Their mean MRR@10, MAP@10, and Hit@10 are also lower.

Table 5.8: BM25+BGE-M3+E5 before and after fine-tuning on Zalo fold 0.

BGE-M3 state	Seed	NDCG	Δ	MRR	MAP	Hit
Pretrained	Fixed	**0.7595**	N/A	**0.7077**	**0.7048**	0.9266
Fine-tuned	42	0.7320	−0.0275	0.6771	0.6743	0.9141
Fine-tuned	43	0.7340	−0.0255	0.6771	0.6747	0.9203
Fine-tuned	44	0.7378	**−0.0217**	0.6794	0.6771	**0.9281**
Fine-tuned	Mean	0.7346	−0.0249	0.6779	0.6754	0.9208

BM25+BGE-M3+E5 also has lower NDCG@10 in all three fine-tuned runs. Its mean decrease is smaller than that of BM25+BGE-M3. Seed 44 has higher Hit@10 than the pretrained baseline but lower NDCG@10 and MRR@10.

Table 5.9 summarizes uncertainty in the NDCG@10 changes. For each query, the mean across the three fine-tuned seeds is compared with its pretrained value. The bootstrap resamples 106 groups of questions linked by legal references. Its tail values, including the Holm-adjusted values, are descriptive bootstrap quantities rather than calibrated hypothesis-test p-values; the procedure is specified in Section 4.8.

Table 5.9: Uncertainty in hybrid NDCG@10 changes on Zalo fold 0.

Comparison	Queries	Groups	Mean Δ	95% group bootstrap interval	Bootstrap tail value	Holm-adjusted tail value
Fine-tuned minus pretrained, BM25+BGE-M3	640	106	−0.0520	[−0.0919, +0.1076]	0.7480	1.0000
Fine-tuned minus pretrained, BM25+BGE-M3+E5	640	106	−0.0249	[−0.0542, +0.0938]	0.7480	1.0000

The intervals in Table 5.9 include both decreases and increases. The observed negative means therefore do not establish a decrease across the wider query population, and the intervals do not establish equivalence. Neither hybrid reaches the predefined +0.01 NDCG@10 improvement threshold. These conclusions apply to the corrected post-hoc comparison with fixed fusion weights.

Fine-Tuning for Standalone Dense Retrieval

Selecting the settings. This experiment selects settings using BGE-M3's standalone dense performance. It evaluates 36 checkpoints across three learning rates, four development folds, and steps 30, 60, and 90. The selection rule requires a mean NDCG@10 gain of at least 0.01, an NDCG@10 increase in at least three of the four folds, and a mean Recall@100 decrease of no more than 0.01.

The Zalo-only training stage is the first to meet this rule, so later data-expansion stages are not run. The selected configuration uses learning rate 2×10^{-5} and the checkpoint at step 60. Table 5.10 reports its development results; NDCG@10 increases in all four folds. These are selected development scores rather than independent test estimates.

Table 5.10: Development results for the selected dense model.

Quantity	**Selected result**
Mean development NDCG@10	0.7504
NDCG@10 gain over the pretrained development baseline	+0.0605
Development MRR@10	0.7111
Development Recall@100	0.9633

Standalone test results. Table 5.11 compares pretrained and fine-tuned BGE-M3 on the 640 official Zalo test queries using extracted article text. This is the official dataset partition, not the grouped fold-0 comparison above. Each experiment uses its own pretrained baseline. The official test queries had been examined before this experiment, so this comparison is also post-hoc.

Table 5.11: BGE-M3 before and after fine-tuning on the official Zalo test set.

System	**NDCG@10**	**Delta NDCG@10**	**MRR@10**	**Recall@100**
Pretrained	0.6740	0.0000	0.6213	**0.9617**
Fine-tuned seed 42	**0.6932**	**0.0192**	**0.6448**	0.9531
Fine-tuned seed 43	0.6925	0.0185	0.6431	0.9516
Fine-tuned seed 44	0.6928	0.0189	0.6437	0.9531
Fine-tuned seed mean	0.6928	0.0189	0.6438	0.9526

All three fine-tuned dense runs have higher NDCG@10 and MRR@10 than pretrained BGE-M3, but lower Recall@100. The mean Recall@100 decrease is 0.0091. These are observed changes on the fixed-corpus test population; no calibrated significance test for this dense comparison is reported here.

Hybrid test results. Table 5.12 evaluates whether the dense model selected above also improves hybrid retrieval. The pretrained and fine-tuned comparisons use the same 640 official Zalo test queries and corpus as the standalone comparison. Fusion weights selected with pretrained BGE-M3 remain fixed: 0.20/0.80 for BM25/BGE-M3 and 0.15/0.50/0.35 for BM25/BGE-M3/E5.

In Table 5.12, relative Δ divides the change by the pretrained metric. Positive-seed counts

indicate runs with a positive NDCG@10 change, rather than runs exceeding the +0.01 improvement threshold.

Table 5.12: Hybrid results after selection by standalone dense performance.

Quantity	**BM25+BGE-M3**	**BM25+BGE-M3+E5**
Pretrained NDCG@10	**0.7526**	0.7467
Fine-tuned mean NDCG@10	0.7483	**0.7476**
Δ NDCG@10	−0.0043	+0.0009
Relative Δ NDCG@10	−0.5707%	+0.1203%
Δ Recall@100	−0.0057	−0.0031
Seeds with positive Δ NDCG@10	0/3	2/3
NDCG@10 improvement threshold	Not met	Not met

Neither hybrid reaches the predefined +0.01 NDCG@10 improvement threshold. BM25+BGE-M3 has lower mean NDCG@10, while BM25+BGE-M3+E5 has a small positive difference. Mean Recall@100 decreases for both configurations. These observations apply to the post-hoc test comparison with fixed fusion weights.

Changes in Individual Queries

The aggregate changes do not apply equally to every question. Table 5.13 reports query-level changes for seed 42 in the hybrid-selected experiment on grouped Zalo fold 0. Table 5.14 reports changes for the standalone-selected experiment on the official Zalo test set, using each query's mean metric across fine-tuned seeds 42, 43, and 44. Both populations contain 640 queries, but the comparisons use their own test allocations and pretrained baselines.

A top-ten miss means that no recorded target is matched in the first ten retrieved articles. In Table 5.13, the first three rows divide all queries into higher, lower, or unchanged NDCG@10. The remaining rows count transitions into or out of the top ten and are not additional groups to sum with those rows. In Table 5.14, Improved, Worse, and Tied refer to positive, negative, and unchanged NDCG@10 differences. Hybrid weights remain fixed.

Table 5.13: Query-level changes after hybrid-based selection.

Outcome	**BM25+BGE-M3**	**BM25+BGE-M3+E5**
Higher NDCG@10	102	102
Lower NDCG@10	210	163
Unchanged NDCG@10	328	375
Previously missing at ten, now retrieved	23	16
Previously retrieved at ten, now missing	48	24

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Outcome	BM25+BGE-M3	BM25+BGE-M3+E5
Missing at ten in both systems	32	31

In the hybrid-selected experiment, BM25+BGE-M3 introduces more than twice as many new top-ten misses as it recovers. BM25+BGE-M3+E5 introduces fewer new misses, but these still outnumber recovered misses. Both hybrids nevertheless improve some queries.

Table 5.14: Query-level changes after standalone dense selection.

Comparison	Improved	Worse	Tied
Dense	127	96	417
BM25+BGE-M3	81	93	466
BM25+BGE-M3+E5	85	86	469

The standalone dense comparison has more improved than worsened queries. Both hybrid comparisons have slightly more worsened than improved queries, with most queries unchanged.

Summary of Fine-Tuning Results

The hybrid-selected configuration increases development scores but lowers test NDCG@10 for both hybrids in every reported seed. The standalone-selected configuration increases dense NDCG@10 and MRR@10 in every seed but lowers Recall@100. Neither corresponding hybrid reaches the predefined +0.01 NDCG@10 improvement threshold.

These post-hoc results show an observed dense ranking gain without a practically useful hybrid gain under the fixed fusion weights. The experiments use different training procedures and test allocations, so their differences cannot be attributed to a single parameter. Retuned fusion weights and transfer to ALQAC or BCA were not evaluated.

5.3.3 Overall Retrieval Benchmark

This section compares the pretrained retrieval methods on the supplied ALQAC and Zalo reference corpora. The BCA results in Section 5.3.1 form the public-service comparison; the fine-tuning results in Section 5.3.2 remain separate because they use different selection procedures and matched test populations.

Benchmark Conditions

The overall benchmark uses only the reference article text supplied with ALQAC and Zalo: 811 queries and 3,363 articles for ALQAC, and 3,196 queries and 61,425 articles

for Zalo. Within each dataset, all methods use the same questions, corpus, and recorded relevance targets. Each query receives equal weight and one out-of-fold prediction under the grouped selection procedure in Section 4.8.

Lexical and Hybrid Retrieval

Table 5.15 compares BM25, pretrained BGE-M3, pretrained multilingual-E5-base (E5), and the hybrid configurations on ALQAC and Zalo. Table 5.16 reports their NDCG@10 changes relative to BM25.

Table 5.15: Pretrained retrieval results on ALQAC and Zalo.

Method	**NDCG@10**	**MRR@10**	**MAP@10**	**Hit@10**
ALQAC (811 queries)				
BM25	0.7986	0.7648	0.7621	0.9112
BGE-M3 pretrained	0.8437	0.8134	0.8108	0.9433
E5 pretrained	0.8126	0.7729	0.7722	0.9371
BM25+BGE-M3	**0.8805**	**0.8534**	**0.8516**	**0.9692**
BM25+BGE-M3+E5	0.8784	0.8519	0.8504	0.9630
Zalo (3,196 queries)				
BM25	0.5264	0.4731	0.4682	0.7100
BGE-M3 pretrained	0.6424	0.5917	0.5850	0.8235
E5 pretrained	0.5640	0.5104	0.5046	0.7509
BM25+BGE-M3	0.6805	0.6298	0.6235	**0.8583**
BM25+BGE-M3+E5	**0.6818**	**0.6317**	**0.6254**	0.8576

BGE-M3 has the highest NDCG@10 among the individual retrievers on both datasets. Both hybrids score higher than BM25 and the individual dense retrievers.

Table 5.16 reports absolute NDCG@10 changes as hybrid minus BM25. Relative changes divide these differences by the corresponding BM25 NDCG@10.

Table 5.16: Hybrid NDCG@10 gains over BM25.

Dataset	**Hybrid**	**Absolute change**	**Relative change**
ALQAC	BM25+BGE-M3	**+0.0819**	**+10.2590%**
ALQAC	BM25+BGE-M3+E5	+0.0798	+9.9970%
Zalo	BM25+BGE-M3	+0.1541	+29.2760%

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Dataset	Hybrid	Absolute change	Relative change
Zalo	BM25+BGE-M3+E5	**+0.1554**	**+29.5210%**

The absolute NDCG@10 gains over BM25 are larger in Zalo than in ALQAC.

Contribution of E5

The contribution of E5 is measured by comparing BM25+BGE-M3+E5 with BM25+BGE-M3 on the same evaluation queries. Absolute changes subtract the BM25+BGE-M3 NDCG@10 value; relative changes divide by that same baseline.

Table 5.17: NDCG@10 changes from adding E5.

Dataset	Absolute change	Relative change	Stored Wilcoxon p
ALQAC	-0.0021	-0.2380%	0.4498
Zalo	+0.0013	+0.1900%	0.1548

Adding E5 slightly reduces NDCG@10 on ALQAC and slightly increases it on Zalo. Table 5.17 reports unadjusted query-level paired Wilcoxon tests. Questions linked by shared legal references may remain dependent, so these tests do not establish variation across independent legal topics.

Neither reported query-level test provides sufficient evidence of an improvement from adding E5. ALQAC has a negative observed difference, while the small positive Zalo difference is not supported as an improvement by its reported test.

Highest-Scoring Retrieval Configurations

On ALQAC, BM25+BGE-M3 has the highest observed NDCG@10, MRR@10, MAP@10, and Hit@10. Adding E5 slightly reduces all four metrics.

On Zalo, BM25+BGE-M3+E5 has the highest NDCG@10, MRR@10, and MAP@10, while BM25+BGE-M3 has the highest Hit@10. The small positive NDCG@10 difference from adding E5 is not supported as an improvement by the reported query-level test.

For the 484-query BCA group, BM25+BGE-M3+E5 has the highest observed NDCG@10 and Hit@10, whereas BM25+BGE-M3 has the highest MRR@10 and MAP@10. These descriptive results do not establish an advantage across independent legal groups.

Across the evaluated conditions, both hybrids have higher NDCG@10 than BM25. The contribution of E5 varies by condition and metric. Overall, the highest-scoring configu-

ration depends on the metric and dataset. These results do not establish a general E5 improvement.

5.3.4 Summary of Results

From the overall benchmark, pretrained BGE-M3 has higher NDCG@10 than BM25 and E5 on both ALQAC and Zalo.

Both hybrids BM25+BGE-M3 and BM25+BGE-M3+E5 have higher NDCG@10 than BM25 and the individual dense retrievers. The contribution of E5 is small and varies by condition.

The finetune version of BGE-M3 does not improve hybrid retrieval in both dataset experiments. This provides evidence that the pretrained BGE-M3 is already well-suited for hybrid retrieval in the legal domain, and further fine-tuning does not yield significant improvements.

Chapter 6

Conclusion and Future Work

This chapter synthesizes the experimental findings rather than repeating the numerical analysis from Chapter 5. The conclusions distinguish improvement over BM25, incremental value beyond an existing hybrid, and transfer from dense adaptation into a complete retrieval system.

6.1 Summary of Findings

6.1.1 Hybrid Retrieval over the Lexical Baseline

The experiments support the value of lexical–semantic hybrid retrieval over the fixed BM25 reference in the evaluated conditions. Both the BM25+BGE-M3 hybrid and the BM25+BGE-M3+E5 hybrid improve NDCG@10 over BM25 on the ALQAC and Zalo reference corpora, and BCA shows the same direction in the reported 484-query group. The result is strongest as an empirical finding about these collections and their specified evaluation populations. It does not imply that a single weight setting, corpus construction policy, or hybrid architecture will transfer unchanged to other legal corpora.

6.1.2 Incremental Contribution of E5

On the reference-text benchmark, adding E5 changes NDCG@10 by approximately -0.0021 on ALQAC and +0.0013 on Zalo. Neither reported query-level test provides sufficient evidence of an improvement. In the 484-query BCA group, adding E5 increases observed NDCG@10 and Hit@10 but decreases MRR@10 and MAP@10; these are descriptive results within one legal-reference group. The reported comparisons therefore do not establish a general improvement from adding E5 to BM25+BGE-M3.

6.1.3 Effect of Legal-Domain Adaptation

The answer is mixed and is one of the main findings of the thesis. Study A: Hybrid-Oriented Adaptation shows substantial development-selection gains, but its corrected hybrid evaluation on data unseen during training and selection shows mean NDCG@10 decreases of 0.0520 for the two-way hybrid and 0.0249 for the three-way hybrid. Study

B: Dense-Oriented Adaptation shows that fine-tuning can improve BGE-M3 as a dense retriever, increasing mean NDCG@10 by 0.0189. However, that improvement largely disappears after the adapted encoder is integrated into locked-weight hybrids. Thus, dense adaptation can improve an encoder without improving the complete lexical–semantic retrieval system by a meaningful amount.

6.2 Research Contributions

The first contribution is a controlled empirical evaluation of lexical–semantic hybrid retrieval for Vietnamese legal information retrieval. The thesis compares BM25, BGE-M3, E5, two-way fusion, and three-way fusion on the supplied ALQAC and Zalo reference corpora and the separately constructed BCA corpus. It separates improvement over a lexical baseline from incremental improvement over an already strong hybrid.

The second contribution is an evaluation of legal-domain BGE-M3 adaptation at both dense-retriever and complete-hybrid levels. The work preserves negative and non-transfer results rather than treating them as failed experiments. The distinction between dense-only improvement and hybrid-system improvement is practically important for legal retrieval systems, where the deployed system is usually a pipeline rather than a single encoder.

The third contribution is a dataset and evaluation contribution involving the BCA collection. The thesis constructs a retrieval collection from public questions and published answer citations, analyzes target coverage, and applies component-aware evaluation. This contribution shows that corpus availability and query dependence can materially affect how legal retrieval metrics should be interpreted.

6.3 Discussion and Limitations

The experiments show that hybrid retrieval is useful, but they also show why retrieval evaluation should not stop at a favorable aggregate score. A hybrid can improve over BM25 while an additional component contributes little. A model can repair some queries while introducing other failures. A benchmark can contain many questions while having fewer independent legal components than the query count suggests. These distinctions matter in legal retrieval because a system must retrieve specific provisions, not merely topically related documents.

The E5 comparison separates a numerical change from supported improvement. On the reference corpora, ALQAC has a small negative difference and Zalo a small positive difference, with neither test establishing an improvement. These results support evaluating E5 against BM25+BGE-M3 rather than attributing the full hybrid gain over BM25 to E5.

The adaptation results illustrate the limits of development selection. Study A's fine-tuning recipe performs well on development queries but regresses in corrected hybrid evaluation on unseen data. Study B shows that this is not simply because legal-domain fine-tuning is incapable of improving dense retrieval. Instead, the complete hybrid can absorb, dilute, or counteract dense improvements because fusion weights, lexical evidence, and score calibration all interact. The present experiments do not isolate which mechanism dominates.

The publicly available Vietnamese legal data suitable for supervised retrieval adaptation remain limited in the present setting. Additional diverse and carefully validated query–article training pairs may improve the fine-tuning process, particularly by providing broader coverage of legal topics and relevance relations. However, the study did not evaluate this intervention; it is therefore a hypothesis for future work rather than an empirical conclusion.

BCA exposes dataset limitations that are not model limitations. Some targets are absent from the indexed corpus, and some answerable queries are only partially covered. A ranking model cannot retrieve evidence that is not present. Therefore, end-to-end scores should be interpreted as measuring the whole retrieval pipeline, including corpus construction and target mapping, not only scoring quality.

Several validity limits remain. The evaluation is restricted to Vietnamese legal collections and recorded relevance judgments. It does not establish multilingual generalization, cross-jurisdiction transfer, downstream legal-answer correctness, current legal validity, user satisfaction, latency, throughput, or deployment cost. Study A's corrected adaptation evaluation on data unseen during training and selection is post-hoc because it follows a flawed preliminary analysis of that same evaluation fold. Study B also uses a fixed-corpus benchmark with prior test observation, so it should not be described as a pristine prospective confirmation.

Statistical conclusions are also bounded. Non-significant intervals do not prove equivalence, and detectable differences do not automatically imply practical importance. The thesis therefore distinguishes observed mean changes, practical improvement thresholds, and statistical uncertainty.

6.4 Future Work

A new, prospectively controlled evaluation should be conducted on an untouched query set. Corpus construction, model selection, fusion configuration, and statistical procedures should be fixed before examining the final test results.

Corpus quality should also be improved, particularly for the BCA collection. Coverage gaps can be addressed through version-aware mapping of legal documents, while query-level

failures can be systematically sampled and annotated using a defined rubric. This would help distinguish missing evidence, annotation limitations, lexical mismatch, semantic confusion, and fusion-related ranking errors.

Future work should further extend the evaluation from retrieval performance to the complete RAG system. This includes assessing answer correctness, citation support, legal sufficiency, latency, throughput, and computational cost. Retrieval metrics remain useful indicators, but they cannot replace direct evaluation of generated legal answers and deployed system performance.

6.5 Closing Remarks

This thesis shows that Vietnamese legal retrieval benefits from lexical–semantic hybridization, but also that more components or better dense-only scores do not automatically produce a better complete system. The most important lesson is methodological: legal retrieval claims must be tied to the exact corpus, target coverage, selection boundary, and system level being evaluated. A disciplined negative result can be as informative as a positive result when it prevents an unsupported deployment claim.

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